

$$2. \quad x^2 + y^2 = 1, y = z$$

OR 2005 Spring Q. No. 7(v)

Find the position vector of the curve $x^2 + y^2 = 1, y = z$ in parametrical representation.

Solution: Given that,

$$x^2 + y^2 = 1, y = z$$

Clearly, the given curve is ellipse in xyz-plane.

Put $x = r \cos t, y = r \sin t$, and $y = z$ with $r = 1$.

So, $x = \cos t, y = \sin t, z = \sin t$

Thus, $(x, y, z) = (\cos t, \sin t, \sin t)$

C. Find a tangent vector and the corresponding unit tangent vector $\vec{u}(t)$ at a given point.

$$1. \quad \vec{r}(t) = t\vec{i} + t^3\vec{j} \text{ at } P(1, 1, 0)$$

Solution: Given that,

$$\vec{r}(t) = t\vec{i} + t^3\vec{j} + 0\vec{k}$$

$$\text{So, } \frac{d\vec{r}}{dt} = \vec{r}'(t) = \vec{i} + 3t^2\vec{j} + 0\vec{k}$$

Therefore, the tangent vector be,

$$\frac{d\vec{r}}{dt} = (1, 3t^2, 0) \text{ at } (1, 1, 0)$$

Since, $\vec{r}(t) = (t, t^3, 0) = (x, y, z)$ at P.

$$\Rightarrow x = t, y = t^3, z = 0 \text{ at P}$$

$$\Rightarrow x = 1, y = 1, z = 0 \text{ at P.}$$

Thus, $t = 1$. So that, $\frac{d\vec{r}}{dt} = (1, 3, 0)$

Also, the unit tangent vector be

$$\hat{r}'(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|} = \frac{(1, 3, 0)}{\sqrt{1^2 + 3^2 + 0^2}} = \left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}, 0\right)$$

$$2. \quad \vec{r}(t) = \cos t \vec{i} + 2\sin t \vec{j} \text{ at } P\left(\frac{1}{2}, \sqrt{3}, 0\right).$$

[2008 Spring Q. No. 7(b)]

Solution: Given that,

$$\vec{r}(t) = \cos t \vec{i} + 2\sin t \vec{j} + 0\vec{k} = (x, y, z)$$

So, $x = \cos t, y = 2\sin t, z = 0$.

And given point be, $P\left(\frac{1}{2}, \sqrt{3}, 0\right)$.

$$\text{So, } \frac{1}{2} = \cos t, \sqrt{3} = 2 \sin t, 0 = 0$$

$$\Rightarrow \cos 60^\circ = \cos t, \frac{\sqrt{3}}{2} = \sin t \text{ i.e. } \sin 60^\circ = \sin t.$$

$$\text{Then, } t = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{\sin 60^\circ}{\cos 60^\circ}\right) = 60^\circ$$

$$\text{Since, } \vec{r}(t) = \cos t \vec{i} + 2\sin t \vec{j}$$

$$\text{So, } \vec{r}'(t) = -\sin t \vec{i} + 2\cos t \vec{j}$$

$$= -\sin 60^\circ \vec{i} + 2\cos 60^\circ \vec{j} \text{ at P}$$

$$= -\frac{\sqrt{3}}{2} \vec{i} + 2 \cdot \frac{1}{2} \vec{j} \text{ at P}$$

$$= \left(-\frac{\sqrt{3}}{2}, 1, 0\right)$$

Thus, tangent vector at P is, $\left(-\frac{\sqrt{3}}{2}, 1, 0\right)$

And, unit tangent vector at P is

$$\begin{aligned} \hat{r}'(t) &= \frac{\vec{r}'(t)}{|\vec{r}'(t)|} = \frac{\left(-\frac{\sqrt{3}}{2}, 1, 0\right)}{\sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + 1^2 + 0^2}} \\ &= \frac{1}{\sqrt{\frac{3}{4} + 1}} \left(-\frac{\sqrt{3}}{2}, 1, 0\right) \\ &= \frac{2}{\sqrt{7}} \left(-\frac{\sqrt{3}}{2}, 1, 0\right) \\ &= \left(-\frac{\sqrt{3}}{\sqrt{7}}, \frac{2}{\sqrt{7}}, 0\right) \end{aligned}$$

$$3. \quad \vec{r}(t) = \cosh t \vec{i} + \sinh t \vec{j} \text{ at } P\left(\frac{5}{3}, \frac{4}{3}, 0\right)$$

Solution: Given that,

$$\vec{r}(t) = \cosh t \vec{i} + \sinh t \vec{j}$$

$$\text{So, } \vec{r}'(t) = \sinh t \vec{i} + \cosh t \vec{j}$$

$$\text{Given point be, } P(x, y, z) = P\left(\frac{5}{3}, \frac{4}{3}, 0\right)$$

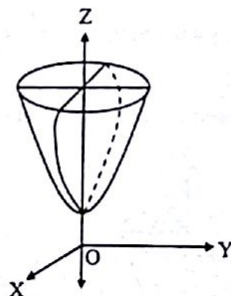
$$\Rightarrow x = \frac{5}{3}, y = \frac{4}{3}, z = 0$$

$$\text{i.e. } \cosh t = \frac{5}{3}, \sinh t = \frac{4}{3}$$

(a) In yz -plane: When $x = 0$ then (i) gives, $z = y^2$. This is a parabola having openward toward the positive z -axis.

(b) In zx -plane: When $y = 0$ then (i) gives, $z = x^2$. This is a parabola having openward toward the positive z -axis.

With the information the sketch of the hyperboloid is as in figure.



2005 Fall Q. No. 7(a)

Define tangent and tangent plane of a curve at a point.

Solution: Tangent to a Curve:

Let C be a curve in space. The tangent on C at a point P of C is the limiting position of a straight line that through P and a point Q of C as Q tends to P along C .

Tangent Plane to a Curve:

Let C be a curve in space. The tangents on C at a point P of C , is called tangent plane.

2006 Spring Q. No. 7(d); 2006 Fall Q. No. 7(e); 2007 Spring Q. No. 7(c)

Write equation of an ellipsoid. Sketch it with centre and axis of symmetry.

Solution: The equation of hyperboloid of two sheet is $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$... (i)

For sketch:

(i) **Centre:** Clearly (i) has centre at $(0, 0, 0)$.

(ii) **Symmetry:** Since all three variables have same degree, so it is symmetrical about all three axes.

(iii) **Intercept:**

When $y = 0 = z$ then $x^2 = a^2$. So, $x = \pm a$. Therefore, the figure cuts x -axis at $x = \pm a$.

When $x = 0 = z$ then $y^2 = b^2$. So, $y = \pm b$. Therefore, the figure cuts y -axis at $y = \pm b$.

When $x = 0 = y$ then $z^2 = c^2$. So, $z = \pm c$. Therefore, the figure cuts z -axis at $z = \pm c$.

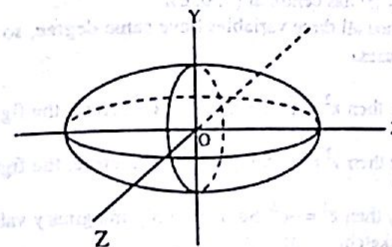
(iv) **Plane section:**

(a) In xy -plane: When $z = 0$ then (i) gives, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. This is an ellipse.

(b) In yz -plane: When $x = 0$ then (i) gives, $\frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. This is an ellipse.

(c) In zx -plane: When $y = 0$ then (i) gives, $\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1$. This is an ellipse.

With the information, the sketch of the ellipsoid is as in figure.



2007 Fall Q. No. 7(e); 2012 Fall Q. No. 7(d)

Find the parametric representation of the surface $x^2 + 4y^2 = 9, z = 3$.

Solution: Given that,

$$x^2 + 4y^2 = 9, z = 3.$$

$$\Rightarrow \frac{x^2}{9} + \frac{y^2}{9/4} = 1, z = 3.$$

Clearly, the given curve is ellipsoid in xyz -plane.

Put $y = t$ then $x = \pm \sqrt{9 - 4t^2}$ and $z = 3$.

Thus, $(x, y, z) = (\pm \sqrt{9 - 4t^2}, t, 3)$.

2011 Fall Q. No. 7(a)

Represent the equation $x^2 + y^2 = 9, z = 5 \tan^{-1}(y/x)$ parametrically.

Solution: Given that,

$$x^2 + y^2 = 9, z = 5 \tan^{-1}(y/x).$$

Clearly, the given curve is ellipsoid in xyz -plane.

Put, $x = r \cos t, y = r \sin t$ with $r = 3$.

Therefore, $z = 5 \tan^{-1}(\sin t / \cos t) = 5t$.

So, $x = 3 \cos t, y = 3 \sin t, z = 5t$.

Thus, $(x, y, z) = (3 \cos t, 3 \sin t, 5t)$.

2011 Fall Q. No. 7(b)

Find the unit tangent vector to the curve $\vec{r}(t) = 2 \cos t \vec{i} + \sin t \vec{j}$ at $(\sqrt{2}, \sqrt{2}, 0)$.

Hint: Similar to exercise 9.1 Q. No. C(ii).

OTHER SOLVED PROBLEMS

- Write the equation of hyperboloid of one sheet and then sketch.

Thus, the tangent vector at P be

$$\vec{r}'(t)_{\text{at } P} = \frac{4}{3}\vec{i} + \frac{5}{3}\vec{j} + 0\vec{k} = \left(\frac{4}{3}, \frac{5}{3}, 0\right)$$

And, unit tangent vector at P be,

$$\begin{aligned}\hat{r}'(t)_{\text{at } P} &= \frac{\left(\frac{4}{3}, \frac{5}{3}, 0\right)}{\sqrt{\left(\frac{4}{3}\right)^2 + \left(\frac{5}{3}\right)^2 + 0^2}} \\ &= \frac{\left(\frac{4}{3}, \frac{5}{3}, 0\right)}{\sqrt{\frac{16}{9} + \frac{25}{9} + 0}} = \frac{3}{\sqrt{41}} \left(\frac{4}{3}, \frac{5}{3}, 0\right) = \left(\frac{4}{\sqrt{41}}, \frac{5}{\sqrt{41}}, 0\right).\end{aligned}$$

OTHER IMPORTANT QUESTION FROM FINAL EXAM

SHORT QUESTIONS

2002 Q. No. 7(d)

Write the equation of hyperboloid and paraboloid.

Solution: Equation of Hyperboloid:

The equation of hyperboloid of one sheet is $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ and of two sheet is $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$.

Equation of Paraboloid:

The equation of paraboloid is $x^2 + y^2 = z$.

2003 Fall Q. No. 7(b)

Write the equation of hyperboloid of two sheet and then sketch.

Solution: The equation of hyperboloid of two sheet is $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$... (i)

For sketch:

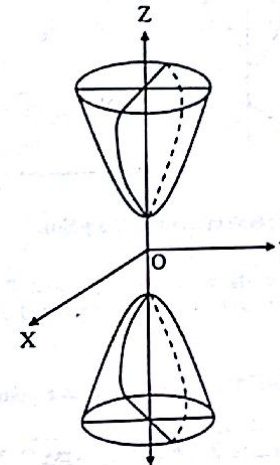
- (i) Centre: Clearly (i) has centre at (0, 0, 0).
- (ii) Symmetry: Since all three variables have same degree, so it is symmetrical about all three axes.
- (iii) Intercept: When $y = 0 = z$ then $x^2 = -a^2$. So, x has only imaginary value which is non-acceptable for sketch.
When $x = 0 = z$ then $y^2 = -b^2$. So, y has only imaginary value which is also non-acceptable for sketch.
When $x = 0 = y$ then $z^2 = c^2$. So, $z = \pm c$. Therefore, the figure cuts z-axis at $z = \pm c$.
- (iv) Plane section:

(a) In xy-plane: When $z = 0$ then (i) gives, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = -1$. This is an imaginary ellipse.

(b) In yz-plane: When $x = 0$ then (i) gives, $\frac{y^2}{b^2} - \frac{z^2}{c^2} = -1 \Rightarrow \frac{z^2}{c^2} - \frac{y^2}{b^2} = 1$.
This is a hyperbola.

(c) In zx-plane: When $y = 0$ then (i) gives, $\frac{x^2}{a^2} - \frac{z^2}{c^2} = -1 \Rightarrow \frac{z^2}{c^2} - \frac{x^2}{a^2} = 1$.
This is a hyperbola.

With the information the sketch of the hyperboloid is as in figure.



2003 Fall Q. No. 7(a)

What do you mean by tangent to a curve?

Solution: Tangent to a Curve:

Let C be a curve in space. The tangent on C at a point P of C is the limiting position of a straight line that through P and a point Q of C as Q tends to P along C.

2005 Fall Q. No. 7(b)

Sketch the paraboloid $z = x^2 + y^2$... (i)

Solution: The equation of paraboloid is $z = x^2 + y^2$

For sketch:

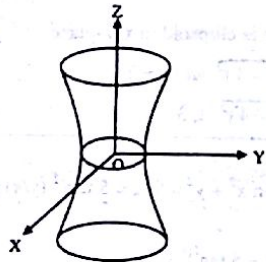
- (i) Symmetry: Since the paraboloid has x and y with degree 2, so it is symmetrical about x-axis and y-axis.
- (ii) Intercept: Clearly the paraboloid has vertex at origin. So, it does not intersect the x-axis and y-axis and z-axis except at origin.
- (iii) Plane section:

Solution: The equation of hyperboloid of two sheet is $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$... (i)

For sketch:

- (i) **Centre:** Clearly (i) has centre at (0, 0, 0).
- (ii) **Symmetry:** Since all three variables have same degree, so it is symmetrical about all three axes.
- (iii) **Intercept:**
When $y = 0 = z$ then $x^2 = a^2$. So, $x = \pm a$. Therefore, the figure cuts x-axis at $x = \pm a$.
When $x = 0 = z$ then $y^2 = b^2$. So, $y = \pm b$. Therefore, the figure cuts y-axis at $y = \pm b$.
When $x = 0 = y$ then $z^2 = -c^2$. So, z has only imaginary value which is non-acceptable for sketch.
- (iv) **Plane section:**
 - (a) **In xy-plane:** When $z = 0$ then (i) gives, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. This is an ellipse.
 - (b) **In yz-plane:** When $x = 0$ then (i) gives, $\frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$. This is a hyperbola.
 - (c) **In zx-plane:** When $y = 0$ then (i) gives, $\frac{x^2}{a^2} - \frac{z^2}{c^2} = 1$. This is a hyperbola.

With the information the sketch of the hyperboloid is as in figure.



2. Write the equation of cone and then sketch.

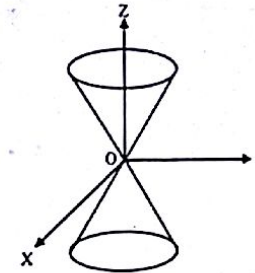
Solution: The equation of hyperboloid of two sheet is $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$... (i)

For sketch:

- (i) **Centre:** Clearly (i) has centre at (0, 0, 0).
- (ii) **Symmetry:** Since all three variables have same degree, so it is symmetrical about all three axes.
- (iii) **Intercept:**
When $y = 0 = z$ then $x^2 = 0$. So, $x = 0$. Therefore, the figure cuts x-axis at origin. And, other two gives same value.
- (iv) **Plane section:**
 - (a) **In xy-plane:** When $z = 0$ then (i) gives, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 0$. This is a point ellipse.
 - (b) **In yz-plane:** When $x = 0$ then (i) gives, $\frac{y^2}{b^2} - \frac{z^2}{c^2} = 0 \Rightarrow y = \pm \frac{bz}{c}$

(c) **In zx-plane:** When $y = 0$ then (i) gives, $\frac{x^2}{a^2} - \frac{z^2}{c^2} = 0 \Rightarrow x = \pm \frac{az}{c}$

With the information the sketch of the hyperboloid is as in figure.



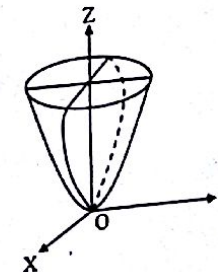
3. Write the equation of elliptic paraboloid and then sketch.

Solution: The equation of hyperboloid of two sheet is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2z}{c}$... (i)

For sketch:

- (i) **Centre:** Clearly (i) has vertex at (0, 0, 0).
- (ii) **Symmetry:** Since both x and y have same degree, so it is symmetrical about both x-axis and y-axis.
- (iii) **Intercept:**
When $y = 0 = z$ then $x^2 = 0$. So, $x = 0$. Therefore, the figure cuts x-axis at origin. And, other two gives same value. Thus, the figure cuts the axes only at origin.
- (iv) **Plane section:**
 - (a) **In xy-plane:** When $z = 0$ then (i) gives, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 0$. This is a point ellipse.
 - (b) **In yz-plane:** When $x = 0$ then (i) gives, $y^2 = \frac{2b^2 z}{c}$.
 - (c) **In zx-plane:** When $y = 0$ then (i) gives, $x^2 = \frac{2a^2 z}{c}$.

With the information the sketch of the hyperboloid is as in figure.



Clearly, $z^{n-1} F(z)$ has poles at $z = -1$ and at $z = -2$ of simple order.

Since we have,

$$z^{-1} \{F(z)\} = \text{sum of residue of } z^{n-1} F(z) \quad \dots\dots(iv)$$

Here, residue of $z^{n-1} F(z)$ at $z = -1$ is

$$\begin{aligned} \text{Res}_{z=-1} z^{n-1} F(z) &= \lim_{z \rightarrow -1} (z+1) z^{n-1} F(z) \\ &= \lim_{z \rightarrow -1} \left(\frac{z^n}{z+2} \right) = \frac{(-1)^n}{1} = (-1)^n \end{aligned}$$

And residue of $z^{n-1} F(z)$ at $z = -2$ is

$$\begin{aligned} \text{Res}_{z=-2} z^{n-1} F(z) &= \lim_{z \rightarrow -2} (z+2) z^{n-1} F(z) \\ &= \lim_{z \rightarrow -2} \left(\frac{z^n}{z+1} \right) = \frac{(-2)^n}{-1} = -(-2)^n \end{aligned}$$

Thus, (iii) becomes with (iv) as

$$y_n = (-1)^n - (-2)^n$$

This is the required solution of (i) satisfying (ii).

(x) $y_{n+2} - 3y_{n+1} + 2y_n = 4^n$ where $y_0 = 0, y_1 = 1$.

[2008 Spring Q. No. 3(b)]

Solution: Given difference equation is

$$y_{n+2} - 3y_{n+1} + 2y_n = 4^n \quad \dots\dots(i)$$

with $y_0 = 0$ and $y_1 = 1$ $\dots\dots(ii)$

Now, taking z-transform on (i) then,

$$Z\{y_{n+2}\} - 3Z\{y_{n+1}\} + 2Z\{y_n\} = Z\{4^n\}$$

Then,

$$z^2 \left(\bar{y} - y_0 - \frac{y_1}{z} \right) - 3z(\bar{y} - y_0) + 2\bar{y} = \frac{z}{z-4}$$

Applying (ii) then,

$$\begin{aligned} \bar{y}(z^2 - 3z + 2) - z &= \frac{z}{z-4} \\ \Rightarrow \bar{y} &= \frac{1}{z^2 - 3z + 2} \left(\frac{z}{z-4} + z \right) \\ &= \frac{z^2 - 3z}{(z-1)(z-2)(z-4)} \end{aligned}$$

And taking inverse z-transform we get,

$$y_n = z^{-1} \left\{ \frac{z^2 - 3z}{(z-1)(z-2)(z-4)} \right\} \quad \dots\dots(iii)$$

Let,

$$F(z) = \frac{z^2 - 3z}{(z-1)(z-2)(z-4)}$$

Then,

$$z^{n-1} F(z) = \frac{z^{n+1} - 3z^n}{(z-1)(z-2)(z-4)}$$

Clearly, $z^{n-1} F(z)$ has poles at $z = 1, z = 2, z = 4$ of simple order.

Since we have,

$$z^{-1} \{F(z)\} = \text{sum of residues of } z^{n-1} F(z) \quad \dots\dots(iv)$$

Here, residue of $z^{n-1} F(z)$ at $z = 1$ is,

$$\begin{aligned} \text{Res}_{z=1} z^{n-1} F(z) &= \lim_{z \rightarrow 1} [(z-1) z^{n-1} F(z)] \\ &= \lim_{z \rightarrow 1} \left(\frac{z^{n+1} - 3z^n}{(z-2)(z-4)} \right) = \frac{1-3}{(-1)(-3)} = \frac{-2}{3} \end{aligned}$$

And residue of $z^{n-1} F(z)$ at $z = 2$ is,

$$\begin{aligned} \text{Res}_{z=2} z^{n-1} F(z) &= \lim_{z \rightarrow 2} (z-2) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 2} \left(\frac{z^{n+1} - 3z^n}{(z-1)(z-4)} \right) = \frac{2^{n+1} - 3(2^n)}{(1)(-2)} = \frac{2^n(2-3)}{-2} = 2^{n-1} \end{aligned}$$

Also, residue of $z^{n-1} F(z)$ at $z = 4$ is

$$\begin{aligned} \text{Res}_{z=4} z^{n-1} F(z) &= \lim_{z \rightarrow 4} (z-4) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 4} (z-4) \left(\frac{z^{n+1} - 3z^n}{(z-1)(z-2)} \right) \\ &= \frac{4^{n+1} - 3(4^n)}{(3)(2)} = \frac{4^n(4-3)}{6} = \frac{4^n}{6} = \frac{2^{2n}}{6} = \frac{2^{2n-1}}{3} \end{aligned}$$

Therefore, (iii) becomes with (iv) is

$$y_n = -\frac{2}{3} + 2^{n-1} + \frac{2^{2n-1}}{3}$$

This is the required solution of (i) satisfying (ii).

(xi) $y_{n+2} - 3y_{n+1} + \frac{1}{4}y_n = 1$ where $y_0 = 1, y_1 = 2$.

(xii) $y_{n+2} - 4y_{n+1} + 4y_n = 2^n$ where $y_0 = 0, y_1 = 1$.

[2011 Spring Q. No. 3(b)] [2012 Fall Q. No. 3(b)]

Solution: Given difference equation is

$$y_{n+2} - 4y_{n+1} + 4y_n = 2^n \quad \dots\dots(i)$$

with $y_0 = 0, y_1 = 1$ $\dots\dots(ii)$

Now, taking z-transform on (i) then

$$Z\{y_{n+2}\} - 4Z\{y_{n+1}\} + 4Z\{y_n\} = Z\{2^n\}$$

Then,

$$z^2 \left(\bar{y} - y_0 - \frac{y_1}{z} \right) - 4z(\bar{y} - y_0) + 4\bar{y} = \frac{z}{z-2} \text{ for } y_n = z\{y_n\}$$

Applying (ii) then,

$$\begin{aligned} z^2 \left(\bar{y} - \frac{1}{z} \right) - 4z\bar{y} + 4\bar{y} &= \frac{z}{z-2} \\ \Rightarrow \bar{y}(z^2 - 4z + 4) &= \frac{z}{z-2} + z \end{aligned}$$

$$z^2 \left(\bar{y} - y_0 - \frac{y_1}{z} \right) - 4z(\bar{y} - y_0) + 4\bar{y} = 0 \quad \text{where, } \bar{y} = Z\{y_k\}$$

Applying (ii) then

$$\begin{aligned} \bar{y}(z^2 - 4z + 4) - z^2 - 4z &= 0 \\ \Rightarrow \bar{y} &= \frac{z^2 - 4z}{(z-2)^2} \end{aligned}$$

And taking inverse z-transform, we get

$$y_k = Z^{-1} \left\{ \frac{z^2 - 4z}{(z-2)^2} \right\} \quad \dots\dots\dots(3)$$

$$\text{Let, } F(z) = \frac{z^2 - 4z}{(z-2)^2}$$

Thus, with [by Q. No. 2(iv)], the equation (3) gives,

$$y_k = 2^k (1 - k)$$

(vii) $y_{k+2} - 4y_{k+1} + 4y_k = 0$ with $y_0 = 1, y_1 = 0$.

Solution: Same question to (vi) with replacing x by k .

(viii) $x_{n+1} - y_n = 1, y_{n+1} + x_n = 0$ where $x_0 = 0, y_0 = -1$.

Solution: Given difference equation is

$$x_{n+1} - y_n = 1 \quad \dots\dots(1)$$

$$y_{n+1} + x_n = 0 \quad \dots\dots(2)$$

$$\text{with } x_0 = 0, y_0 = -1 \quad \dots\dots(3)$$

Now, taking z-transform on (1) and (2) then,

$$Z\{x_{n+1}\} - Z\{y_n\} = Z\{1\}$$

$$Z\{y_{n+1}\} + Z\{x_n\} = Z\{0\}$$

Then,

$$z(\bar{x} - x_0) - \bar{y} = z/(z-1) \quad \text{for } \bar{x} = Z\{x_n\}, \bar{y} = Z\{y_n\}$$

$$z(\bar{y} - y_0) + \bar{x} = 0$$

Applying (3) then,

$$z\bar{x} - \bar{y} = z/(z-1) \quad \dots\dots(4)$$

$$z\bar{y} + \bar{x} = -z \quad \dots\dots(5)$$

Substituting the value of $\bar{y}$ from (4) in (5) then,

$$z\left(z\bar{x} - \frac{z}{z-1}\right) + \bar{x} = -z$$

$$\Rightarrow \bar{x}(z^2 + 1) = -z + \frac{z^2}{z-1} = \frac{z}{z-1}$$

$$\Rightarrow \bar{x} = \frac{z}{(z-1)(z^2+1)} \quad \dots\dots\dots(6)$$

And taking inverse z-transform on (vi) then,

$$x_n = z^{-1} \left\{ \frac{z}{(z-1)(z^2+1)} \right\} \quad \dots\dots\dots(7)$$

$$\text{Let, } F(z) = \frac{z}{(z-1)(z^2+1)}$$

Thus, with [by Q. No. 2(i)], the equation (3) gives,

$$x_n = \frac{1}{2} \left[\frac{1}{2} - \left(\cos\left(\frac{n\pi}{2}\right) + \sin\left(\frac{n\pi}{2}\right) \right) \right]$$

Replacing $n+1$ by n then,

$$\begin{aligned} y_n &= -\frac{1}{2} \left[1 - \left(\cos\left(\frac{(n-1)\pi}{2}\right) + \sin\left(\frac{(n-1)\pi}{2}\right) \right) \right] \\ &= -\frac{1}{2} \left[1 - \left(\cos\frac{n\pi}{2} \cos\frac{\pi}{2} + \sin\frac{n\pi}{2} \sin\frac{\pi}{2} + \sin\frac{n\pi}{2} \cos\frac{\pi}{2} - \cos\frac{n\pi}{2} \sin\frac{\pi}{2} \right) \right] \\ &= -\frac{1}{2} \left[1 - \left(\sin\left(\frac{n\pi}{2}\right) - \cos\left(\frac{n\pi}{2}\right) \right) \right] \end{aligned}$$

Thus,

$$x_n = \frac{1}{2} \left[1 - \cos\left(\frac{n\pi}{2}\right) - \sin\left(\frac{n\pi}{2}\right) \right] \quad \text{and} \quad y_n = \frac{1}{2} \left[-1 - \cos\left(\frac{n\pi}{2}\right) + \sin\left(\frac{n\pi}{2}\right) \right]$$

be solution of (i) and (ii) satisfying (iii).

(ix) $y_{n+2} + 3y_{n+1} + 2y_n = 0$ where $y_0 = 0, y_1 = 1$.

Solution: Given difference equation is

$$y_{n+2} + 3y_{n+1} + 2y_n = 0 \quad \dots\dots(i)$$

$$\text{with } y_0 = 0, y_1 = 1 \quad \dots\dots(ii)$$

Now, taking Z-transform on (i) then,

$$Z\{y_{n+2}\} + 3Z\{y_{n+1}\} + 2Z\{y_n\} = Z\{0\}$$

Then,

$$z^2 \left(\bar{y} - y_0 - \frac{y_1}{z} \right) + 3z\bar{y} + 2\bar{y} = 0 \quad \text{for } \bar{y} = Z\{y_n\}$$

Applying (ii) then,

$$z^2 \left(\bar{y} - \frac{1}{z} \right) + 3z\bar{y} + 2\bar{y} = 0$$

$$\Rightarrow \bar{y}(z^2 + 3z + 2) = z$$

$$\Rightarrow \bar{y} = \frac{z}{z^2 + 3z + 2}$$

Now, taking inverse z-transform we get,

$$y_n = z^{-1} \left\{ \frac{z}{z^2 + 3z + 2} \right\} \quad \dots\dots\dots(iii)$$

$$\text{Let, } F(z) = \frac{z}{z^2 + 3z + 2} = \frac{z}{(z+2)(z+1)}$$

$$\text{Then } z^{n-1} F(z) = \frac{z^n}{(z+1)(z+2)}$$

Also residue of $z^{n-1} F(z)$ at $z = -3$ is,

$$\begin{aligned}\text{Res}_{z=-3} z^{n-1} F(z) &= \lim_{z \rightarrow -3} (z+3) z^{n-1} F(z) \\ &= \lim_{z \rightarrow -3} \left(\frac{z^{n+2} - 2z^{n+1} + 3z^n}{(z-1)^2 (z+1)} \right) \\ &= \frac{(-3)^{n+2} - 2(-3)^{n+1} + 3(-3)^n}{(-4)^2 (-2)} \\ &= \frac{(-3)^n [9 + 6 + 3]}{-32} = \frac{(-3)^n (18)}{-32} = -\frac{(-3)^n \cdot 9}{16}\end{aligned}$$

Thus, with (iv), equation (iii) becomes,

$$z^{-1} \{F(z)\} = \frac{4n-3}{16} + \frac{3(-1)^n}{4} - \frac{9(-3)^n}{16}$$

This is the required solution of (i) satisfying (ii).

(xiv) $2y_n - 2y_{n-1} + y_{n-2} = u(n)$ where $y_n = 0$ for $n < 0$ and $u(n) = 1$ for $n \geq 0$, $u(n) = 0$ for $n < 0$.

Solution: Given difference equation is,

$$2y_n - 2y_{n-1} + y_{n-2} = u(n) \quad \dots\dots(i)$$

Now, taking Z-transform on (i) then

$$\begin{aligned}2\bar{y} - 2z^{-1}\bar{y} + z^{-2}\bar{y} &= Z\{u(n)\} \\ \Rightarrow \bar{y} (2 - 2z^{-1} + z^{-2}) &= \frac{z}{z-1} \\ \Rightarrow \bar{y} \left(\frac{2z^2 - 2z + 1}{z^2} \right) &= \frac{z}{z-1} \\ \Rightarrow \bar{y} &= \frac{z^3}{(z-1)(2z^2 - 2z + 1)}\end{aligned}$$

And taking inverse z-transform we get,

$$y_n = Z^{-1} \left\{ \frac{z^3}{(z-1)(2z^2 - 2z + 1)} \right\} \quad \dots\dots(ii)$$

Let,

$$F(z) = \frac{1}{2} \frac{z^3}{(z-1)(z^2 - z + 1/2)}$$

$$\begin{aligned}\text{Then, } z^{n-1} F(z) &= \frac{z^{n+2}}{2(z-1)(z^2 - z + 1/2)} \\ &= \frac{z^{n+2}}{2(z-1) \left(z - \left(\frac{1}{2} + \frac{i}{2} \right) \right) \left(z - \left(\frac{1}{2} - \frac{i}{2} \right) \right)}\end{aligned}$$

Clearly, $z^{n-1} F(z)$ has poles at $z = 1$, $z = (1+i)/2$, $z = (1-i)/2$.

Since we have,

$$Z^{-1} \{F(z)\} = \text{sum of residue of } z^{n-1} F(z) \quad \dots\dots(iii)$$

Here, residue of $z^{n-1} F(z)$ at $z = 1$ is,

$$\begin{aligned}\text{Res}_{z=1} z^{n-1} F(z) &= \lim_{z \rightarrow 1} (z-1) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 1} \left(\frac{1}{2} \right) \left(\frac{z^{n+2}}{z^2 - z + 1/2} \right) \\ &= \left(\frac{1}{2} \right) \left(\frac{1}{1-1+1/2} \right) \\ &= 1\end{aligned}$$

And, residue of $z^{n-1} F(z)$ at $z = (1+i)/2$ is,

$$\begin{aligned}\text{Res}_{z=(1+i)/2} z^{n-1} F(z) &= \lim_{z \rightarrow (1+i)/2} (z - (1+i)/2) z^{n-1} F(z) \\ &= \lim_{z \rightarrow (1+i)/2} \frac{z^{n+2}}{2(z-1) \left(z - \left(\frac{1}{2} - \frac{i}{2} \right) \right)} \\ &= \left(\frac{1}{2} \right) \frac{(1+i)^{n+2}}{2^{n+2} \left(\frac{1+i}{2} - 1 \right) \left(\frac{1+i}{2} - \left(\frac{1}{2} - \frac{i}{2} \right) \right)} \\ &= \left(\frac{1}{2} \right) \frac{(1+i)^{n+2}}{2^{n+2} \left(\frac{1-i}{2} \right) i}\end{aligned}$$

Also, residue of $z^{n-1} F(z)$ at $z = (1-i)/2$ is,

$$\begin{aligned}\text{Res}_{z=(1-i)/2} z^{n-1} F(z) &= \left(\frac{1}{2} \right) \frac{(1-i)^{n+2}}{2^{n+2} \left(\frac{-i-1}{2} \right) (-i)} \\ &= \left(\frac{1}{2} \right) \frac{(1-i)^{n+2}}{2^{n+2} \left(\frac{i+1}{2} \right) (i)}\end{aligned}$$

Thus,

$$y_n = Z^{-1} \{F(z)\} = 2\pi i \left(1 + \left(\frac{1}{2} \right) \frac{(1+i)^{n+2}}{2^{n+2} \left(\frac{1-i}{2} \right) i} + \left(\frac{1}{2} \right) \frac{(1-i)^{n+2}}{2^{n+2} \left(\frac{i+1}{2} \right) (i)} \right)$$

(xv) $y_n + \frac{y_{n-1}}{4} = \delta(n) + \frac{\delta(n-1)}{3}$ for $n \geq 0$, $y_0 = 0$.

Solution: Given difference equation is,

$$y_n + \frac{y_{n-1}}{4} = \delta(n) + \frac{\delta(n-1)}{3} \quad \dots\dots(i)$$

Now, taking z-transform on (i) then

$$Z\{y_n\} + \frac{1}{4} Z\{y_{n-1}\} = Z\{\delta(n)\} + \frac{1}{3} Z\{\delta(n-1)\}$$

Then,

$$\bar{y} + \frac{1}{4} z^{-1} \bar{y} = 1 + \frac{1}{3} z^{-1}$$

$$\Rightarrow \bar{y} = \frac{z^2 - z}{(z-2)(z-2)^2}$$

$$\Rightarrow \bar{y} = \frac{z^2 - z}{(z-2)^3}$$

And taking inverse z-transform we get,

$$y_n = z^{-1} \left\{ \frac{z^2 - z}{(z-2)^3} \right\} \dots\dots\dots(\text{iii})$$

$$\text{Let, } F(z) = \frac{z^2 - z}{(z-2)^3}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^{n+1} - z^n}{(z-2)^3}$$

Clearly $z^{n-1} F(z)$ has poles at $z = 2$ of order 3.

Since we have,

$$z^{-1} \{F(z)\} = \text{sum of residues of } z^{n-1} F(z) \dots\dots\dots(\text{iv})$$

Here, residue of $z^{n-1} F(z)$ at $z = 2$ of order 3 is,

$$\begin{aligned} \text{Res}_{z=2} z^{n-1} F(z) &= \lim_{z \rightarrow 2} \left(\frac{1}{2} \frac{d^2}{dz^2} [(z-2)^3 z^{n-1} F(z)] \right) \\ &= \lim_{z \rightarrow 2} \left(\frac{1}{2} \frac{d^2}{dz^2} (z^{n+1} - z^n) \right) \\ &= \lim_{z \rightarrow 2} \left(\frac{1}{2} \frac{d}{dz} [(n+1)z^n - nz^{n-1}] \right) \\ &= \lim_{z \rightarrow 2} \left[\frac{(n+1)nz^{n-1} - n(n-1)z^{n-2}}{2} \right] \\ &= \frac{1}{2} (n(n+1)2^{n-1} - n(n-1)2^{n-1}) \\ &= n2^{n-3} [2n + 2 - n + 1] \\ &= n2^{n-3} (n+3) \\ &= n(n+3)2^{n-3} \end{aligned}$$

Thus, with (iv), the equation (iii) becomes,

$$y_n = n(n+3)2^{n-3}$$

This is the required solution of (i) satisfying (ii).

(xiii) $y_{n+2} + 4y_{n+1} + 3y_n = 2n$ where $y_0 = 0, y_1 = 1$.

Solution: Given difference equation is,

$$y_{n+2} + 4y_{n+1} + 3y_n = 2n \dots\dots(\text{i})$$

$$\text{with } y_0 = 0, y_1 = 1 \dots\dots(\text{ii})$$

Now, taking z-transform on (i) then

$$Z\{y_{n+2}\} + 4Z\{y_{n+1}\} + 3Z\{y_n\} = 2Z\{n\}$$

$$\text{Then, } z^2 \left(\bar{y} - y_0 - \frac{y_1}{z} \right) + 4z(\bar{y} - y_0) + 3\bar{y} = 2 \left(\frac{z}{(z-1)^2} \right)$$

Applying (ii) then,

$$\bar{y}(z^2 + 4z + 3) - z = \frac{2z}{(z-1)^2}$$

$$\begin{aligned} \Rightarrow \bar{y} &= \left(\frac{1}{z^2 + 4z + 3} \right) \left(\frac{2z}{(z-1)^2} + z \right) \\ &= \frac{z^3 - 2z^2 + 3z}{(z-1)^2(z+3)(z+1)} \end{aligned}$$

And taking inverse z-transform we get,

$$y_n = z^{-1} \left\{ \frac{z^3 - 2z^2 + 3z}{(z-1)^2(z+1)(z+3)} \right\} \dots\dots\dots(\text{iii})$$

Let,

$$F(z) = \frac{z^3 - 2z^2 + 3z}{(z-1)^2(z+1)(z+3)}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^{n+2} - 2z^{n+1} + 3z^n}{(z-1)^2(z+1)(z+3)}$$

Clearly $z^{n-1} F(z)$ has poles at $z = 1$ of order 2 and at $z = -1, z = -3$ of simple order.

Since we have,

$$z^{-1} \{F(z)\} = \text{sum of residues of } z^{n-1} F(z) \dots\dots\dots(\text{iv})$$

Here, residue of $z^{n-1} F(z)$ at $z = 1$ of order 2 is,

$$\begin{aligned} \text{Res}_{z=1} z^{n-1} F(z) &= \lim_{z \rightarrow 1} \frac{d}{dz} \left(\frac{z^{n+2} - 2z^{n+1} + 3z^n}{(z+1)(z+3)} \right) \\ &= \lim_{z \rightarrow 1} \frac{d}{dz} \left(\frac{z^{n+2} - 2z^{n+1} + 3z^n}{z^2 + 4z + 3} \right) \\ &= \lim_{z \rightarrow 1} \frac{(z^2 + 4z + 3) [(n+2)(z)^{n+1} - 2(n+1)z^n + 3nz^{n-1}] - 2(n+1)z^n + 3nz^{n-1}}{(z^2 + 4z + 3)^2} \\ &= \frac{(1+4+3) [(n+2) - 2(n+1) + 3n] - (1-2+3)(2+4)}{(1+4+3)^2} \\ &= \frac{8(2n) - 12}{64} \\ &= \frac{4n-3}{16} \end{aligned}$$

And residue of $z^{n-1} F(z)$ at $z = -1$ is,

$$\begin{aligned} \text{Res}_{z=-1} z^{n-1} F(z) &= \lim_{z \rightarrow -1} (z+1) z^{n-1} F(z) \\ &= \lim_{z \rightarrow -1} \left(\frac{z^{n+2} - 2z^{n+1} + 3z^n}{(z-1)^2(z+3)} \right) \\ &= \frac{(-1)^{n+2} - 2(-1)^{n+1} + 3(-1)^n}{(-2)^2(2)} \\ &= \frac{(-1)^n (1+2+3)}{8} \\ &= \frac{3(-1)^n}{4} \end{aligned}$$

$$\Rightarrow z^2 \left[\bar{y} - y_0 - \frac{y_1}{z} \right] + (a+b) z [\bar{y} - y_0] + ab \bar{y} = 0$$

$$\Rightarrow \bar{y} (z^2 + (a+b)z + ab) = z^2 y_0 + zy_1 + (a+b) zy_0$$

$$\Rightarrow \bar{y} = \frac{z^2 y_0 + zy_1 + (a+b) zy_0}{z^2 + (a+b)z + ab} = F(z) \text{ (let)}$$

Then we have, $y_n = z^{-1} (F(z))$.

And we have,

$$z^{-1} (F(z)) = \text{sum of residues of } z^{n-1} F(z).$$

Now,

$$z^{n-1} F(z) = \frac{z^{n+1} y_0 + z^n y_1 + (a+b) z^n y_0}{(z+a)(z+b)}$$

Here poles are $z = -a, -b$ which are simple.

So,

$$\begin{aligned} \text{Res}_{z=-a} z^{n-1} F(z) &= \frac{(-a)^{n+1} y_0 + (-a)^n y_1 + (a+b) (-a)^n y_0}{b-a} \\ &= \frac{(-a)^{n+1} y_0 + (-a)^n y_1 - (a)^{n+1} y_0 + b(a)^n y_0}{b-a} \\ &= \frac{by_0 + y_1}{b-a} (-a)^n \end{aligned}$$

And,

$$\begin{aligned} \text{Res}_{z=-b} z^{n-1} F(z) &= \frac{(-b)^{n+1} y_0 + (-b)^n y_1 + (a+b) (-b)^n y_0}{a-b} \\ &= \frac{(-b)^{n+1} y_0 + (-b)^n y_1 - a(-b)^n y_0 - (b)^{n+1} y_0}{a-b} \\ &= \frac{ay_0 + y_1}{a-b} (-b)^n \end{aligned}$$

Now,

$$\begin{aligned} y_n &= z^{-1} (F(z)) \\ \Rightarrow y_n &= \frac{by_0 + y_1}{b-a} (-a)^n + \frac{ay_0 + y_1}{a-b} (-b)^n \end{aligned}$$

5. Solve the difference equation

$$y_{n+2} + 2ay_{n+1} + a^2 y_n = 0$$

Solution: Given difference equation is,

$$y_{n+2} + 2ay_{n+1} + a^2 y_n = 0.$$

Now, taking z-transform on (i) then,

$$Z\{y_{n+2}\} + 2a Z\{y_{n+1}\} + a^2 Z\{y_n\} = 0$$

$$\Rightarrow z^2 \left[\bar{y} - y_0 - \frac{y_1}{z} \right] + 2az (\bar{y} - y_0) + a^2 \bar{y} = 0$$

$$\Rightarrow \bar{y} (z^2 + 2az + a^2) = z^2 y_0 + zy_1 + 2azy_0$$

$$\Rightarrow \bar{y} = (z^2 y_0 + zy_1 + 2azy_0)/(z+a)^2 = F(z)$$

Then we have, $y_n = z^{-1} (F(z))$.

And we have,

$$z^{-1} (F(z)) = \text{sum of residues of } z^{n-1} F(z).$$

Now,

$$z^{n-1} F(z) = (z^{n+1} y_0 + z^n y_1 + 2az^n y_0)/(z+a)^2$$

Here $z^{n-1} F(z)$ has pole at $z = -a$ of degree 2.

$$\begin{aligned} \text{Res}_{z=-a} z^{n-1} F(z) &= \lim_{z \rightarrow -a} \frac{d}{dz} \left((z+a)^2 \left(\frac{z^{n+1} y_0 + z^n y_1 + 2az^n}{(z+a)^2} \right) y_0 \right) \\ &= \lim_{z \rightarrow -a} [(n+1) z^n y_0 + n z^{n-1} y_1 + 2a n z^{n-1} y_0] \\ &= (n+1) (-a)^n y_0 + n(-a)^{n-1} y_1 + 2an (-a)^{n-1} y_0 \\ &= (-a)^n y_0 + n(-a)^{n-1} [y_1 + 2ay_0 - ay_0] \\ &= (-a)^n y_0 + [ay_0 + y_1] n (-a)^{n-1} \end{aligned}$$

Now,

$$\begin{aligned} y_n &= z^{-1} (F(z)) \\ \Rightarrow y_n &= (-a)^n y_0 + [ay_0 + y_1] n (-a)^{n-1} \end{aligned}$$

6. Consider the difference equation

$$y_{n+2} = y_{n+1} + y_n, \quad \text{where } y_0 = 0, y_1 = 1$$

Find the solution of y_n . Show that the limiting value of $\frac{y_{n+1}}{y_n} = \frac{1+\sqrt{5}}{2}$, when $n \rightarrow \infty$.

Solution: Now, given difference equation is

$$y_{n+2} - y_{n+1} - y_n = 0$$

Taking z-transform both sides,

$$z^2 \left[\bar{y} - y_0 - \frac{y_1}{z} \right] - z (\bar{y} - y_0) - \bar{y} = 0$$

$$\Rightarrow z^2 \bar{y} - z - z \bar{y} - \bar{y} = 0$$

$$\Rightarrow \bar{y} = \frac{z}{z^2 - z - 1} = F(z)$$

Then we have, $y_n = z^{-1} (F(z))$.

And we have,

$$z^{-1} (F(z)) = \text{sum of residues of } z^{n-1} F(z).$$

Now,

$$y_n = z^{-1} (F(z)) = Z^{-1} \left\{ \frac{z}{z^2 - z - 1} \right\}$$

$$\text{Here, } z^{n-1} F(z) = \frac{z^n}{z^2 - z - 1}$$

$$\text{Here poles of } z^{n-1} F(z) \text{ are at } z = \frac{1+\sqrt{1+4}}{2} = \frac{1+\sqrt{5}}{2}$$

which are simple.

$$\Rightarrow \bar{y} \left(\frac{4z+1}{4z} \right) = \left(\frac{3z+1}{3z} \right)$$

$$\Rightarrow \bar{y} = \frac{4}{3} \left(\frac{3z+1}{4z+1} \right) = \frac{z+1/3}{z+1/4}$$

And taking inverse z-transform we get,

$$y_n = z^{-1} \left\{ \frac{z+1/3}{z+1/4} \right\} \dots\dots\dots(ii)$$

Let,

$$F(z) = \frac{z+1/3}{z+1/4}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^{n-1}(z+1/3)}{z+1/4}$$

Clearly, $z^{n-1} F(z)$ has poles at $z = -\frac{1}{4}$

Since we have,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z) \dots\dots\dots(iii)$$

Here, residue of $z^{n-1} F(z)$ at $z = -\frac{1}{4}$ is,

$$\begin{aligned} \text{Res}_{z=-1/4} z^{n-1} F(z) &= \lim_{z \rightarrow -1/4} (z + 1/4) z^{n-1} F(z) \\ &= \lim_{z \rightarrow -1/4} [z^{n-1} (z + 1/3)] \\ &= \left(-\frac{1}{4}\right)^{n-1} \left(-\frac{1}{4} + \frac{1}{3}\right) = \left(-\frac{1}{4}\right)^{n-1} \left(\frac{1}{12}\right) \end{aligned}$$

$$\text{Thus, } y_n = \frac{1}{12} \left(-\frac{1}{4}\right)^{n-1}$$

This is the required solution of (i) satisfying (ii).

$$(xvi) \ y_{n+1} + \frac{y_n}{4} = \left(\frac{1}{4}\right)^{-n} \text{ where } y_0 = 0.$$

Solution: Given difference equation is,

$$y_{n+1} + \frac{y_n}{4} = \left(\frac{1}{4}\right)^{-n} \dots\dots\dots(i)$$

$$\text{with } y_0 = 0 \dots\dots\dots(ii)$$

Now, taking z-transform on (i) then,

$$Z\{y_{n+1}\} + \frac{1}{4} Z\{y_n\} = Z\{4^n\}$$

$$\text{Then, } z(\bar{y} - y_0) + \frac{1}{4} \bar{y} = \frac{z}{z-4}$$

Applying (ii) then

$$\bar{y} \left(z + \frac{1}{4} \right) = \frac{z}{z-4} \Rightarrow \bar{y} = \frac{4z}{(z-4)(4z+1)}$$

$$\Rightarrow \bar{y} = \frac{z}{(z-4)(z+1/4)}$$

And taking inverse z-transform we get,

$$y_n = z^{-1} \left\{ \frac{z}{(z-4)(z+1/4)} \right\} \dots\dots\dots(iii)$$

Let,

$$F(z) = \frac{z}{(z-4)(z+1/4)}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^n}{(z-4)(z+1/4)}$$

Clearly $z^{n-1} F(z)$ has simple poles at $z = 4$, $z = -\frac{1}{4}$.

Since we have,

$$z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z) \dots\dots\dots(iv)$$

Here residue of $z^{n-1} F(z)$ at $z = 4$ is,

$$\begin{aligned} \text{Res}_{z=4} z^{n-1} F(z) &= \lim_{z \rightarrow 4} (z-4) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 4} \left(\frac{z^n}{z+1/4} \right) = \frac{4^n}{4+1/4} = \frac{4^{n+1}}{17} \end{aligned}$$

And residue of $z^{n-1} F(z)$ at $z = -\frac{1}{4}$ is,

$$\begin{aligned} \text{Res}_{z=-1/4} z^{n-1} F(z) &= \lim_{z \rightarrow -1/4} (z + 1/4) z^{n-1} F(z) \\ &= \lim_{z \rightarrow -1/4} \left(\frac{z^n}{z-4} \right) \\ &= \left[\frac{(-1/4)^n}{-1/4-4} \right] = (-1)^{n-1} \left(\frac{1/4}{1/4+4} \right) = (-1)^{n-1} \left(\frac{1}{17} \right) \end{aligned}$$

Thus, with (iv), equation (iii) gives,

$$y_n = \frac{4^{n+1}}{17} + \frac{(-1)^{n-1}}{17}$$

Note: If we reform the question as $y_{n+1} + \frac{y_n}{4} = \left(\frac{1}{4}\right)^n$ with $y_0 = 0$ then we obtain

$$y_n = 2 \left(\frac{1}{4}\right)^n - 2 \left(-\frac{1}{4}\right)^n$$

4. Solve the difference equation

$$y_{n+2} + (a+b)y_{n+1} + aby_n = 0, \text{ where } a \text{ and } b \text{ are constants.}$$

Solution: Given difference equation is,

$$y_{n+2} + (a+b)y_{n+1} + aby_n = 0, \text{ where } a \text{ and } b \text{ are constants}$$

Now, taking z-transform on (i) then,

$$Z\{y_{n+2}\} + (a+b)Z\{y_{n+1}\} + abZ\{y_n\} = 0$$

2009 Spring Q. No. 2(b)

Define Z transform. State and prove linearity property of z transform. Using it find Z transform of $\cos at$ and then find Z transform of $e^{at} \cos bt$.

2009 Fall Q. No. 5(b)

Define Z-transform of a function $f(t)$ and by using the definition find the Z-transform of (i) $(-1)^n$ and (ii) n

2009 Fall Q. No. 6(a)

State the first shifting theorem for Z transformation and hence find $Z[e^{-at}]$.

2011 Spring Q. No. 3.a.

Define Z-transform. State and prove linearity property of Z transform. Find $Z(e^{-iat})$ and hence deduce Z transform of $\sin at$ and $\cos at$.

2011 Fall Q. No. 6(a)

Find the Z-transform of the function $f(t) = e^{-iat}$ and hence deduce the value of $Z(\cos at)$ and $Z(\sin at)$.

2012 Fall Q. No. 6(b)

Define Z-transform of a function $f(t)$ and by using the definition find the Z-transform of (i) $(-1)^n$ (ii) $\frac{1}{n!}$.

Z transform Inverse**2002 Q. No. 3(a)**

$$\text{Find } Z^{-1} \left[\frac{z}{z^2 + 7z + 10} \right]$$

2004 Spring Q. No. 3(a); 2007 Spring Q. No. 3(b)

$$\text{Find the inverse Z-transform of the function } \frac{2z}{(z-1)(z^2+1)}$$

2004 Fall Q. No. 3(a); 2009 Fall Q. No. 6(b)

$$\text{Find the inverse Z-transform of the function } \frac{z+2}{z^2-5z+6}$$

2005 Fall Q. No. 3(a)

Show that $Z[nf(t)] = -z \frac{d}{dz} [F(z)]$ where $F(z) = Z[f(t)]$, and find

$$Z^{-1} \left[\frac{z}{(z+1)^2(z-1)} \right]$$

2006 Spring Q. No. 2(b) OR

$$\text{Find the inverse Z-transform of } \frac{z}{z^2-2z+2}$$

2007 Fall Q. No. 3(a)

$$\text{Find } z^{-1} \left[\frac{z^2-3}{(z+2)(z^2+1)} \right]$$

2008 Spring Q. No. 3(a)

State and prove initial and final value theorem on Z transform. Evaluate

$$Z^{-1} \left(\frac{z}{(z-1)(z-2)} \right).$$

2008 Fall Q. No. 6(b)

$$\text{Find inverse Z transform of } \frac{z^2+1}{z^2-2z+2}$$

2009 Spring Q. No. 3(a)

If $Z\{f(t)\} = f(z)$, derive the expression of $Z\{f(t + kT)\}$. Evaluate

$$Z^{-1} \left[\frac{z+3}{(z+1)(z+2)} \right].$$

2011 Fall Q. No. 4(a)

Define Z-transform? Find the inverse of the Z-transform of the function:

$$\frac{2z}{(z-1)(z^2+1)}$$

Application of Z transform**2002 Q. No. 3(b)**

Solve the difference equation $y_{n+1} + y_n = 1$ given $y_0 = 0$

2004 Fall Q. No. 3(b)

Solve the difference equation $y_{n+1} + 3y_n = 5^n$ given $y_0 = 0$.

2005 Fall Q. No. 3(b)

Show that $Z(y_{n+k}) = z^k \left[\bar{y} - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2} - \dots - \frac{y_{k-1}}{z^{k-1}} \right]$ where $\bar{y} = Z(y_n)$. and then

solve the differential equation $y_{n+2} = 1$, where $y_0 = 0 = y_1$.

2007 Fall Q. No. 3(b)

Solve the difference equation $y_{n+2} - 8y_{n+1} + 16y_n = 4^n$, where $y_0 = 0$ and $y_1 = 1$; by unit z-transform.

2011 Fall Q. No. 5(a)

Using Z-transform, solve the difference equation $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ when $y_0 = y_1 = 0$.

$$\text{Res}_{z = \frac{1+\sqrt{5}}{2}} z^{n-1} F(z) = \lim_{z \rightarrow \frac{1+\sqrt{5}}{2}} \frac{z^n}{\left(z - \left(\frac{1-\sqrt{5}}{2}\right)\right)} = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n}{\sqrt{5}}$$

And,

$$\text{Res}_{z = \frac{1-\sqrt{5}}{2}} z^{n-1} F(z) = \lim_{z \rightarrow \frac{1-\sqrt{5}}{2}} \frac{z^n}{\left(z - \left(\frac{1+\sqrt{5}}{2}\right)\right)} = \frac{\left(\frac{1-\sqrt{5}}{2}\right)^n}{-\sqrt{5}}$$

Here, $y_n = z^{-1} (F(z))$

$$\Rightarrow y_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n \right]$$

Again, we know

$$y_{n+1} = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1} \right]$$

Now,

$$\lim_{n \rightarrow \infty} \frac{y_{n+1}}{y_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1} \right]}{\frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n \right]}$$

Here the term $\frac{1-\sqrt{5}}{2} = -0.618 < 1$

So when $n \rightarrow \infty$, $(-0.618)^n \rightarrow 0$

So, we get

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{y_{n+1}}{y_n} &= \lim_{n \rightarrow \infty} \frac{\left(\frac{1+\sqrt{5}}{2}\right)^{n+1}}{\left(\frac{1+\sqrt{5}}{2}\right)^n} = \lim_{n \rightarrow \infty} \left(\frac{1+\sqrt{5}}{2}\right) \\ \Rightarrow \lim_{n \rightarrow \infty} \frac{y_{n+1}}{y_n} &= \frac{1+\sqrt{5}}{2} \end{aligned}$$

OTHER IMPORTANT QUESTION FROM FINAL EXAM FOR PRACTICE

2002 Q. No. 3(b) OR

Find Z-transform of $\cosh at \sin bt$.

2002 Q. No. 2(b)

Find $Z[\sin at]$ and $Z[\cos at]$

2003 Fall Q. No. 4(a); 2007 Spring Q. No. 2(b)

Define Z-transform of a function. Find Z-transform of e^{-at} and hence deduce the values of $Z(\cos at)$ and $Z(\sin at)$

2004 Spring Q. No. 2(b)

Define Z-transform of the function. Find (i) $Z(1)$ (ii) $Z(na^n)$ (iii) $Z(e^{-at})$

2004 Fall Q. No. 2(b)

Find Z-transform of $\cos \frac{n\pi}{2}$ and $\sin \frac{n\pi}{2}$

2005 Spring Q. No. 2(b)

Define z-transform. State and prove linearity property of z-transform. Evaluate $Z(t e^{2t})$.

2005 Spring Q. No. 3(a)

Show that $Z[n f(t)] = -z \frac{d}{dz} [Z(f(t))]$. Using it evaluate $Z(nt)$ and $Z(n^2 t)$.

2005 Fall Q. No. 2(b)

Define z-transform. State and prove first shifting theorem of z-transform; and then evaluate $Z(e^{-t})$, $Z(te^{-t})$ and $Z(e^{-t}/n!)$.

2006 Spring Q. No. 2(b)

Define Z-transform. State and prove first shifting theorem of Z-transform. Using it evaluate $Z(te^{2t})$ and $Z(ne^{-2t})$.

2006 Fall Q. No. 4(a)

State and prove linearity property of z-transform. Find $Z(t^k)$, where k is a positive integer.

2006 Fall Q. No. 4(b)

Define convolution of function in z-transform. State and prove convolution theorem in z-transform.

2007 Spring Q. No. 3(a); 2008 Fall Q. No. 6(a); 2012 Fall Q. No. 3(a)

State the first shifting theorem for Z transform and hence find

(i) $Z(\cos at)$ and (ii) $Z(\sin at)$

2007 Fall Q. No. 2(b)

Define z-transform of the function $f(t)$. Find z-transform of $f(t) = n^2 a^n$ and $f(t) = \cos n\theta$.

2008 Spring Q. No. 2(b)

Define Z transform. State and prove first shifting theorem of Z transform. Using it evaluate Z transform of $a^n \cos bt$ and $a^n \sin bt$.

2008 Fall Q. No. 5(b)

Define Z-transform of a function $f(t)$ and by using the definition find the Z-transform of: (i) na^n and (ii) $\frac{1}{n+1}$

$$\text{And } F'' = \frac{\partial^2 F}{\partial r^2} = \frac{\partial}{\partial s} \left(\frac{\partial F}{\partial r} \right) \frac{\partial s}{\partial r} = k^2 \frac{\partial^2 F}{\partial s^2}$$

Then (8) becomes,

$$k^2 \frac{\partial^2 F}{\partial s^2} + \frac{1}{s} \frac{\partial F}{\partial s} + F = 0 \quad \dots\dots\dots (10)$$

This is Bessel's equation with $n = 0$.

Clearly, (10) has solution J_0 and Y_0 where J_0 and Y_0 are Bessel's functions of first and second kind, respectively.

Since, at $n = 0$, Y_0 has no finite value. That means Y_0 has no meaningful solution as $f(r)$ is finite in (5).

Therefore,

$$F(r) = J_0(s) = J_0(kr)$$

Be the solution of (10).

At boundary, $F(R) = 0$, being $G \neq 0$ and by (4)

So, s has infinite roots. Say,

$$s = \alpha_m \quad \text{for } m = 1, 2, 3, \dots\dots\dots$$

$$\Rightarrow k_m = \frac{\alpha_m}{R} \quad \text{for } m = 1, 2, 3, \dots\dots\dots \text{ being } s = kR$$

$$\text{Thus, } F_m(r) = J_0 \left(\frac{r \alpha_m}{R} \right) \text{ for } m = 1, 2, \dots\dots\dots$$

Hence, (6) becomes,

$$u(r, t) = \sum_{m=1}^{\infty} (A_m \cos \lambda_m t + B_m \sin \lambda_m t) + J_0 \left(\frac{r \alpha_m}{R} \right) \quad \dots\dots\dots (11)$$

By (5), $u = (r, 0) = F(r)$ then, (11) gives,

$$f(r) = \sum_{m=1}^{\infty} A_m J_0 \left(\frac{r \alpha_m}{R} \right)$$

Which is Fourier Bessel series. So,

$$A_m = \frac{2}{R^2 J_1^2(\alpha_m)} \int_0^R r g(r) J_0 \left(\frac{r \alpha_m}{R} \right) dr \quad \dots\dots\dots (13)$$

for $m = 1, 2, \dots\dots\dots$

Thus, (11) is required solution with the value of A_m and B_m given in (12) and (13).

Potential function and its solution by spherical membrane

Potential function:

If the Laplacian form, is equal to zero then it called potential equation. That is the equation $\nabla^2 u = 0$ is called potential equation. And the solution of the equation is known as potential function.

Solution: Let $x = r \cos \theta \sin \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \phi$

Then,

$$\nabla^2 u = \frac{1}{r^2} \left[\frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{\sin \phi} \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial u}{\partial \phi} \right) + \frac{1}{\sin^2 \phi} \frac{\partial^2 u}{\partial \theta^2} \right] = 0 \quad \dots\dots\dots (1)$$

Consider the boundary condition, $u(R, \theta, \phi) = f(\phi)$

Then, $u_{\theta\theta} = 0$ as temperature is fixed.

So that (1) becomes,

$$\frac{1}{r^2} \left[\frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{\sin \phi} \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial u}{\partial \phi} \right) \right] = 0 \quad \dots\dots\dots (2)$$

At infinity, the potential will be zero, that is

$$\Delta z \xrightarrow{\infty} u(r, \phi) = 0 \quad \dots\dots\dots (3)$$

Suppose that $u(r, \phi) = A(r) B(\phi)$ (4)

Be solution of (2). Then we get,

$$\begin{aligned} B \frac{\partial}{\partial r} \left(r^2 \frac{\partial A}{\partial r} \right) + \frac{A}{\sin \phi} \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial B}{\partial \phi} \right) &= 0 \\ \Rightarrow \frac{1}{A} \frac{\partial}{\partial r} \left(r^2 \frac{\partial A}{\partial r} \right) &= - \frac{1}{B \sin \phi} \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial B}{\partial \phi} \right) = k \text{ (say)} \end{aligned}$$

This gives,

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial A}{\partial r} \right) = kA \quad \dots\dots\dots (5)$$

And,

$$\frac{1}{\sin \phi} \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial B}{\partial \phi} \right) + kB = 0 \quad \dots\dots\dots (6)$$

Set $\cos \phi = 2$ the, $1 - w^2 = \sin^2 \phi$.

Also,

$$\frac{d}{dQ} = \frac{d}{dw} \left(\frac{dw}{d\phi} \right) = -\sin \phi \frac{d}{dw}$$

So that (6) becomes,

$$\begin{aligned} \frac{d}{dw} \left[(1 - w^2) \frac{dB}{dw} \right] + n(n+1)B &= 0 \quad \text{for } k = n(n+1) \\ \Rightarrow (1 - w^2) \frac{d^2 B}{dw^2} - 2w \frac{dB}{dw} + n(n+1)B &= 0 \end{aligned}$$

Which is Legendre's equation that has solution,

$$B = P_n(w) = P_n(\cos \phi) \quad \text{for } n \geq 0$$

Also, (5) becomes

$$r^2 A'' + 2rA' - n(n+1)A = 0$$

which is called Euler-cauchy equation.

set $A = r^\alpha$ then,

$$\begin{aligned} \alpha(\alpha-1)r^\alpha + 2\alpha r^\alpha - n(n+1)r^\alpha &= 0 \\ \Rightarrow \alpha(\alpha-1) + 2\alpha - n(n+1) &= 0 \\ \Rightarrow \alpha(\alpha+1) - n(n+1) &= 0 \end{aligned}$$

This gives $\alpha = n$ and $\alpha = -(n+1)$

Thus, $A_n = r^n$ and $A_n^* = r^{-(n+1)}$

Hence, (4) becomes,

$$u(r, \phi) = AB$$

$$\Rightarrow u_n(r, \phi) = c_n r^n P_n(\cos \phi) \text{ and } u_n^*(r, \phi) = D_n r^{-(n+1)} P_n(\cos \phi)$$

So that,

$$u(r, \phi) = \sum_{n=0}^{\infty} c_n r^n P_n(\cos \phi) \text{ and } u(r, \phi) = \sum_{n=0}^{\infty} D_n r^{-(n+1)} P_n(\cos \phi)$$

This is required solution.

Exercise 7.4

1. Find the deflection $u(x, y, t)$ of the square membrane with $a = b = 1$ and $c = 1$. If the initial velocity is zero and the initial deflection is

(a) $0.1 \sin 3\pi x \sin 4\pi y$ (b) $kxy(1-x)(1-y)$

Solution: Given that, $a = b = c = 1$.

(a) $g(x, y) = 0$, $f(x, y) = 0.1 \sin 3\pi x \sin 4\pi y$ [2004 Spring Q. No. 4(b) OR]

We have the solution of two dimensional wave equation is

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} (B_{mn} \cos \lambda_{mn} t + B_{mn}^* \sin \lambda_{mn} t) \sin \frac{m\pi}{a} x \sin \frac{n\pi}{b} y \dots (i)$$

where $B_{mn}^* = 0$, being $g(x, y) = 0$

Now,

$$\begin{aligned} B_{mn} &= \frac{4}{ab} \int_0^b \int_0^a f(x, y) \sin \frac{m\pi}{a} x \sin \frac{n\pi}{b} y \, dx \, dy \\ &= 4 \int_0^1 \int_0^1 0.1 \sin 3\pi x \sin 4\pi y \sin m\pi x \sin n\pi y \, dx \, dy \\ &= 0.1 \left[\int_0^1 2 \sin 3\pi x \sin m\pi x \, dx \right] \left[\int_0^1 2 \sin 4\pi y \sin n\pi y \, dy \right] \\ &= 0.1 \int_0^1 [\cos(3-m)\pi x - \cos(3+m)\pi x] \, dx \\ &\quad \int_0^1 [\cos(4-n)\pi y - \cos(4+n)\pi y] \, dy \\ &= 0.1 \left[\frac{\sin(3-m)\pi x}{(3-m)\pi} - \frac{\sin(3+m)\pi x}{(3+m)\pi} \right]_0^1 \left[\frac{\sin(4-n)\pi y}{(4-n)\pi} - \frac{\sin(4+n)\pi y}{(4+n)\pi} \right]_0^1 \\ &= 0.1 \left[\left\{ \frac{\sin(3-m)\pi}{(3-m)\pi} - \frac{\sin(3+m)\pi}{(3+m)\pi} \right\} \left\{ \frac{\sin(4-n)\pi}{(4-n)\pi} - \frac{\sin(4+n)\pi}{(4+n)\pi} \right\} \right] \end{aligned}$$

$$= 0.1 \left(\frac{\sin(3-m)\pi}{(3-m)\pi} \right) \left(\frac{\sin(4-n)\pi}{(4-n)\pi} \right)$$

Therefore,

$$B_{mn} = \frac{0.1}{\pi^2} \left(\frac{\sin(3-m)\pi}{(3-m)} \cdot \frac{\sin(4-n)\pi}{(4-n)} \right) \text{ for } m, n = 1, 2, 3, 4, \dots$$

Now, for B_{34} the term has 0/0 form. So, using L'hospital rule then

$$B_{34} = 0.1 \text{ and others zero.}$$

Now,

$$\lambda_{34} = c\pi \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}} = c\pi \sqrt{\frac{3^2}{1^2} + \frac{4^2}{1^2}} = 5\pi = 5\pi.$$

Thus, the required solution from Equation (i) is

$$\begin{aligned} u(x, y, t) &= B_{34} \cos \lambda_{34} t \sin 3\pi x \sin 4\pi y \\ &= 0.1 \cos 5\pi t \sin 3\pi x \sin 4\pi y \end{aligned}$$

(b) $kxy(1-x)(1-y)$

Solution: Given,

$$\begin{aligned} f(x, y) &= kxy(1-x)(1-y) \\ &= k(x-x^2)(y-y^2) \end{aligned}$$

And, $g(x, y) = 0$

The solution of 2-D wave eqⁿ. is

$$\begin{aligned} u(x, y, t) &= \frac{4}{ab} \int_0^b \int_0^a f(x, y) \sin m\pi x \sin n\pi y \, dx \, dy \\ &= 4 \left[\int_0^1 \int_0^1 f(x, y) \sin m\pi x \sin n\pi y \, dx \, dy \right] \\ &= 4k \left[\int_0^1 (x-x^2) \sin m\pi x \, dx \right] \left[\int_0^1 (y-y^2) \sin n\pi y \, dy \right] \\ &= 4k \left[(x-x^2) \cdot \frac{-\cos m\pi x}{m\pi} - (1-2x) \cdot \frac{-\sin m\pi x}{m^2\pi^2} + (-2) \cdot \frac{\cos m\pi x}{m^3\pi^3} \right]_0^1 \\ &\quad \left[(y-y^2) \cdot \frac{-\cos n\pi y}{n\pi} - (1-2y) \cdot \frac{-\sin n\pi y}{n^2\pi^2} + (-2) \cdot \frac{\cos n\pi y}{n^3\pi^3} \right]_0^1 \\ &= 4k \left(\frac{-\sin m\pi}{m^3\pi^3} - \frac{2\cos m\pi}{m^3\pi^3} + \frac{2}{m^3\pi^3} \right) \left(\frac{-\sin n\pi}{n^3\pi^3} - \frac{2\cos n\pi}{n^3\pi^3} + \frac{2}{n^3\pi^3} \right) \end{aligned}$$

For any value of m , $\sin m\pi = 0$ [$\because m = 1, 2, 3, \dots$]

Then,

where, $\lambda = \alpha$

The equation (6) is called Helmholtz equation.

Let, $F(x, y) = X(x) Y(y)$ (7)

be solution of (6). Then, (6) becomes,

$$Y \frac{\partial^2 X}{\partial x^2} + X \frac{\partial^2 Y}{\partial y^2} + \alpha^2 XY = 0$$

$$\Rightarrow \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = -\frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} - \alpha^2 = -k^2 \text{ (say)}$$

For otherwise the solution is not possible.

That gives,

$$\frac{1}{X} \frac{\partial^2 X}{\partial x^2} = -k^2 \quad \text{and} \quad -\frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} - \alpha^2 = -k^2$$

$$\Rightarrow \frac{\partial^2 X}{\partial x^2} + k^2 X = 0 \quad \text{and} \quad \frac{\partial^2 Y}{\partial y^2} + p^2 Y = 0$$

with $p^2 = \alpha^2 - k^2$

whose solution be

$$X(x) = A \cos kx + B \sin kx$$

$$\text{And} \quad Y(y) = C \cos py + D \sin py$$

where, A, B, C, D are constants.

Then, (7) becomes,

$$F(x, y) = (A \cos kx + B \sin kx) (C \cos py + D \sin py) \quad \text{..... (8)}$$

Since, $u = 0$, for all t . So, $F = 0$ at $x = 0, a$; $y = 0, b$

As $x = 0$ then $A = 0$. Also, as $y = 0$ then $C = 0$.

So that, at $x = a$,

$$B \sin ka = 0$$

$$\Rightarrow \sin ka = 0 \quad \text{as } B \neq 0 \text{ otherwise it leads impossible solution.}$$

$$= \sin n\pi$$

$$\Rightarrow k = \frac{n\pi}{a}$$

Similarly, at $y = b$, $p = \frac{m\pi}{b}$ as $\sin m\pi = 0$

Thus, (8) becomes,

$$F_{mn}(x, y) = A_{mn} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) \quad \text{where, } A_{mn} = B \cdot D$$

be required solution of (6)

And, we have $p^2 = \alpha^2 - k^2$ and $\lambda = c\alpha$

So, $\lambda = c\alpha = c\sqrt{p^2 + k^2}$

$$\Rightarrow \lambda_{mn} = \lambda = c\pi = \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}} \quad [\because \text{using value of } p \text{ and } k]$$

Therefore, the solution of (5) be,

$$G_{mn}(t) = B_{mn} \cos \lambda_{mn} t + B_{mn}^* \sin \lambda_{mn} t$$

Thus, (4) becomes,

$$u(x, y, t) = \sum u_{mn}(x, y, t)$$

$$= \sum F_{mn}(x, y) G_{mn}(t)$$

$$= \sum (B_{mn} \cos \lambda_{mn} t + B_{mn}^* \sin \lambda_{mn} t) \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

As the constants includes A_{mn} i.e. $u(x, y, t)$

$$= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} (B_{mn} \cos \lambda_{mn} t + B_{mn}^* \sin \lambda_{mn} t) \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \quad \text{... (10)}$$

This is required solution of (1)

We have, $u(x, y, 0) = f(x, y)$. Then, (10) gives,

$$f(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right)$$

which is half range multiple Fourier sine series. So,

$$B_{mn} = \frac{2}{a} \frac{2}{b} \int_0^a \int_0^b f(x, y) \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) dy dx$$

$$\Rightarrow B_{mn} = \frac{4}{ab} \int_0^a \int_0^b f(x, y) \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) dy dx \quad \text{..... (11)}$$

Also, we have, $u_t(x, y, 0) = g(x, y)$ then, (10) gives,

$$g(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn}^* \lambda_{mn} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right)$$

which is half range multiple Fourier sine series. So,

$$B_{mn}^* = \frac{4}{ab\lambda_{mn}} \int_0^a \int_0^b g(x, y) \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) dy dx \quad \text{..... (12)}$$

Hence, (10) is required solution of (1) with the coefficients value given in (11) and (12).

Laplacian in polar form

We have, the Laplacian in Cartesian coordinates be,

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \quad \text{..... (1)}$$

Set $x = r \cos \theta$, $y = r \sin \theta$

Then,

$$\frac{\partial u}{\partial x} = u_x = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x}$$

$$= u_r r_x + u_\theta \theta_x$$

And,

$$\frac{\partial^2 u}{\partial x^2} = u_{xx} = u_{rr} r_x^2 + u_{r\theta} r_x \theta_x + u_{\theta r} \theta_x r_x + u_{\theta\theta} \theta_x^2$$

Similarly,

$$\frac{\partial^2 u}{\partial y^2} = u_{yy} = u_{rr} r^2 + u_{r\theta} r + u_{\theta\theta} \theta^2 + u_{\theta r} \theta$$

Since, we have

$$x = r \cos \theta, \quad y = r \sin \theta$$

So, $r^2 = x^2 + y^2$ and $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

Therefore,

$$r_x = \frac{x}{r}, \quad r_{xx} = \left(\frac{x}{r}\right)_x = \frac{r - x \cdot x/r}{r^2} = \frac{r^2 - x^2}{r^3}$$

i.e. $r_x = \frac{x}{r}, \quad r_{xx} = \frac{r^2 - x^2}{r^3}$

Also,

$$r_y = \frac{y}{r} \quad \text{and} \quad r_{yy} = \frac{r^2 - y^2}{r^3}$$

And,

$$\theta_x = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \left(-\frac{y}{x^2}\right) = -\frac{y}{x^2 + y^2} = -\frac{y}{r^2}$$

So, $\theta_{xx} = \left(-\frac{y}{r^2}\right)_x = -y \left(\frac{-2}{r^3} \cdot r_x\right) = \frac{2y}{r^3} \cdot \frac{x}{r} = \frac{2xy}{r^4}$

Similarly,

$$\theta_y = \frac{x}{r^2} \quad \text{and} \quad \theta_{yy} = \frac{-2xy}{r^4}$$

Now,

$$\frac{\partial^2 u}{\partial x^2} = u_{rr} \left(\frac{x}{r}\right)^2 + u_r \left(\frac{y^2}{r^3}\right) + u_{\theta\theta} \left(\frac{y^2}{r^3}\right) + u_{\theta r} \left(\frac{-2xy}{r^4}\right)$$

And,

$$\frac{\partial^2 u}{\partial y^2} = u_{rr} \left(\frac{y}{r}\right)^2 + u_r \left(\frac{x^2}{r^3}\right) + u_{\theta\theta} \left(\frac{x^2}{r^3}\right) + u_{\theta r} \left(\frac{-2xy}{r^4}\right)$$

Thus,

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= u_{rr} \left(\frac{x^2 + y^2}{r^2}\right) + u_r \left(\frac{x^2 + y^2}{r^3}\right) + u_{\theta\theta} \left(\frac{x^2 + y^2}{r^3}\right) + u_{\theta r} \cdot 0 \\ &= u_{rr} \left(\frac{r^2}{r^2}\right) + u_r \left(\frac{r^2}{r^3}\right) + u_{\theta\theta} \left(\frac{r^2}{r^3}\right) \\ &= u_{rr} + \frac{1}{r} u_r + u_{\theta\theta} \cdot \frac{1}{r^2} \end{aligned}$$

This is required form.

Note: The Laplace equation is polar form be,

$$u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = 0$$

being, $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$

Note: The Laplacian in polar form in three dimensional geometry be,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} + u_{zz}$$

Being $x = r \cos \theta, \quad y = r \sin \theta \quad \text{and} \quad z = z$
Such form is also known as in cylindrical coordinate form.

Solution of two dimensional wave equation in circular membrane under certain conditions

Consider two dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{\partial^2 u}{\partial \theta^2} \right) \quad \dots\dots\dots (1)$$

In polar form,

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} \right) \quad \dots\dots\dots (2)$$

Suppose that the circular membrane has radius B and is radially symmetrical function $u(r, t)$. That means u does not depends up on θ . So that

$$\frac{\partial^2 u}{\partial \theta^2} = 0$$

Then (2) becomes,

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right) \quad \dots\dots\dots (3)$$

Consider (3) satisfies the conditions

$$u(R, t) = 0, \text{ for all } t \geq 0. \quad \dots\dots\dots (4)$$

$$u(r, 0) = f(r), \quad u_t(r, 0) = g(r) \quad \dots\dots\dots (5)$$

Let,

$$u(r, t) = F(r) G(t) \quad \dots\dots\dots (6)$$

be the solution of (3) then (6) becomes,

$$F \ddot{G} = c^2 \left(G F'' + \frac{1}{r} G F' \right)$$

This gives,

$$\frac{\ddot{G}}{c^2 G} = \frac{F''}{F} + \frac{F'}{r F} = -k^2 \text{ (say)}$$

That gives,

$$\ddot{G} + \lambda^2 G = 0 \quad \dots\dots\dots (7)$$

And $F'' + \frac{F'}{r} + k^2 F = 0 \quad \dots\dots\dots (8)$

where, $\lambda = ck$

Since (7) is second order differential equation whose solution be,

$$G_m(t) = A_m \cos \lambda_m t + B_m \sin \lambda_m t \quad \dots\dots\dots (9)$$

And, for (8), set $\frac{1}{r} = \frac{k}{s}$. So that,

$$F = \frac{\partial F}{\partial r} = \frac{\partial F}{\partial s} \frac{\partial s}{\partial r} = k \frac{\partial F}{\partial s}$$

$$\begin{aligned}
&= \sum_{n=0}^{\infty} [f(nT+T) - f(nT)] \quad [\because z^n \rightarrow 1 \text{ as } z \rightarrow 1] \\
&= \lim_{n \rightarrow \infty} [f(T) + f(2T) + f(3T) + \dots - f(T) - f(2T) - \dots] \\
&= \lim_{n \rightarrow \infty} [f(n+1)T] - f(0) \\
&= f(\infty) - f(0) \\
\Rightarrow \lim_{z \rightarrow 1} (z-1) F(z) - f(0) &= \lim_{t \rightarrow \infty} f(t) - f(0) \\
\Rightarrow \lim_{z \rightarrow 1} (z-1) F(z) &= \lim_{t \rightarrow \infty} f(t).
\end{aligned}$$

Convolution of functions:

Let $f(t)$ and $g(t)$ are any two functions. Then, the convolution of the functions is denoted by $f * g$ and defined as

$$(f * g)(n) = \sum_{k=0}^{\infty} f(kT) g[(n-k)T]$$

Convolution Theorem

If $F(z)$ and $G(z)$ are Z-transform of $f(t)$ and $g(t)$ respectively then $Z[f(t) * g(t)] = F(z) G(z)$.

Proof: Let, $F(z)$ and $G(z)$ are Z-transform of $f(t)$ and $g(t)$ respectively. Then,

$$\begin{aligned}
F(z) G(z) &= \sum_{n=0}^{\infty} f(nT) z^{-n} \sum_{m=0}^{\infty} g(mT) z^{-m} \\
&= [f(0) + f(T)z^{-1} + f(2T)z^{-2} + \dots] [g(0) + g(T)z^{-1} + g(2T)z^{-2} + \dots] \\
&= [f(0)g(0) + \{f(0)g(T) + f(T)g(0)\}z^{-1} + \{f(0)g(2T) + f(T)g(T) + f(2T)g(0)\}z^{-2} + \dots] \\
&= \sum_{n=0}^{\infty} \left[\sum_{k=0}^{\infty} f(kT)g[(n-k)T] \right] z^{-n} \\
&= \sum_{n=0}^{\infty} (f * g)(nT) z^{-n} \\
&= Z[f * g]
\end{aligned}$$

Thus, $Z[f(t) * g(t)] = F(z) G(z)$.

Exercise 8.1**1. Obtain the Z-transform of**

(i) $Z(e^n)$

Solution: Since we have,

$$\begin{aligned}
Z[a^n f(t)] &= Z[F(x/a)] \quad \text{for } F(z) = Z[f(t)] \\
&= (z[F(x)])_{x \rightarrow x/a}
\end{aligned}$$

Now,

$$Z(e^n) = (Z(1))_{x \rightarrow x/e} = \left(\frac{z}{z-1} \right)_{x \rightarrow x/e} = \frac{z/e}{z/e - 1} = \frac{z}{z-e}$$

(ii) n^3

Solution: Since we have,

$$Z(n a^n) = \frac{az^2}{z(z-a)^2}$$

Now,

$$Z(n^3) = \left[\frac{3z^2}{z(z-3)^2} \right] = \frac{3z}{(z-3)^2}$$

(iii) $r^n \cos n\theta$ (iv) $r^n \sin n\theta$

Solution: Since we have,

$$Z(a^n) = \frac{z}{z-a} \quad \text{and} \quad Z(nf(t)) = -z \frac{d}{dz} (Z[f(t)])$$

Now,

$$\begin{aligned}
Z(re^{in\theta}) &= \frac{z}{z - re^{i\theta}} = \frac{z}{z - r \cos \theta - i r \sin \theta} \times \frac{z - r \cos \theta + i r \sin \theta}{z - r \cos \theta + i r \sin \theta} \\
&= \frac{z(z - r \cos \theta) + i z r \sin \theta}{(z - r \cos \theta)^2 + r^2 \sin^2 \theta} \\
&= \frac{z(z - r \cos \theta) + i z r \sin \theta}{z^2 - 2 r z \cos \theta + 1}
\end{aligned}$$

$$\Rightarrow z[r^n (\cos n\theta + i \sin n\theta)] = \frac{z(z - r \cos \theta)}{z^2 - 2 r z \cos \theta + 1} + i \frac{z r \sin \theta}{z^2 - 2 r z \cos \theta + 1}$$

Comparing real and imaginary part then,

$$Z(r^n \cos n\theta) = \frac{z(z - r \cos \theta)}{z^2 - 2 r z \cos \theta + 1} \quad \text{and} \quad Z(r^n \sin n\theta) = \frac{z r \sin \theta}{z^2 - 2 r z \cos \theta + 1}$$

(v) n

Solution: Since we have,

$$Z(n f(t)) = -z \frac{d}{dz} (Z[f(t)]) \quad \text{and} \quad Z(1) = \frac{z}{z-1}$$

Now,

$$\begin{aligned}
Z(n) &= -z \frac{d}{dz} \left(\frac{z}{z-1} \right) \\
&= -z \frac{d}{dz} \left(\frac{z}{z-1} \right) = -z \left[\frac{z-1-z}{(z-1)^2} \right] = \frac{z}{(z-1)^2}
\end{aligned}$$

(vi) $\frac{1}{n+2}$

Solution: Let $f(t) = \frac{1}{n+2}$

Now,

$$Z(f(t)) = \sum_{n=0}^{\infty} \left(\frac{1}{n+2} \right) z^{-n}$$

$$\begin{aligned}
 &= \frac{1}{2} + \frac{1}{3z} + \frac{1}{4z^2} + \frac{1}{5z^3} + \dots \\
 &= z^2 \left[\frac{1}{2z^2} + \frac{1}{3z^3} + \frac{1}{4z^4} + \frac{1}{5z^5} + \dots \right] + z - z \\
 &= z^2 \left[\frac{1}{z} + \frac{1}{2z^2} + \frac{1}{3z^3} + \frac{1}{4z^4} + \frac{1}{5z^5} + \dots \right] - z \\
 &= -z^2 \log(1 - 1/z) - z \\
 &= -z - z^2 \log(z - 1/z)
 \end{aligned}$$

(vii) $\left\{ \frac{1 - e^{-bt}}{b} \right\}$

Solution: Since we have,

$$Z(1) = \frac{z}{z-1} \quad \text{and} \quad Z\{e^{at} f(t)\} = (Z\{f(t)\})_{z \rightarrow ze^{-aT}}$$

Now,

$$\begin{aligned}
 Z\left\{ \frac{1 - e^{-bt}}{b} \right\} &= -\frac{1}{b} [Z(1) - Z\{e^{-bt}\}] \\
 &= \frac{1}{b} \left[\frac{z}{z-1} - (z(1))_{z \rightarrow ze^{bT}} \right] \\
 &= \frac{1}{b} \left[\frac{z}{z-1} - \left(\frac{z}{z-1} \right)_{z \rightarrow ze^{bT}} \right] \\
 &= \frac{1}{b} \left[\frac{z}{z-1} - \frac{ze^{bT}}{ze^{bT}-1} \right] \\
 &= \frac{1}{b} \left[\frac{z}{z-1} - \frac{z}{z - e^{-bT}} \right]
 \end{aligned}$$

(viii) $t^2 e^{-bt}$

Solution: Since we have,

$$Z(t^k) = -z T \frac{d}{dz} (Z(t^{k-1}))$$

$$Z\{e^{at} f(t)\} = (Z\{f(t)\})_{z \rightarrow ze^{-aT}} \quad \text{and} \quad Z(1) = \frac{z}{z-1}$$

Now,

$$\begin{aligned}
 Z\{t^2 e^{-bt}\} &= (Z\{t^2\})_{z \rightarrow ze^{-aT}} \\
 &= \left(-z T \frac{d}{dz} z(t) \right)_{z \rightarrow ze^{-aT}} \\
 &= \left[-z T \frac{d}{dz} \left\{ -z T \frac{d}{dz} (z(1)) \right\} \right]_{z \rightarrow ze^{-aT}} \\
 &= \left[-z T \frac{d}{dz} \left\{ -z T \frac{d}{dz} \left(\frac{z}{z-1} \right) \right\} \right]_{z \rightarrow ze^{-aT}}
 \end{aligned}$$

$$\begin{aligned}
 &= \left[-z T^2 \frac{d}{dz} \left\{ -z \frac{d}{dz} \left(\frac{1}{(z-1)^2} \right) \right\} \right]_{z \rightarrow ze^{-aT}} \\
 &= \left[-z T^2 \left(\frac{(z-1)^2 - 2z(z-1)}{(z-1)^4} \right) \right]_{z \rightarrow ze^{-aT}} \\
 &= T^2 \left[-z \left(\frac{z-1-2z}{(z-1)^3} \right) \right]_{z \rightarrow ze^{-aT}} \\
 &= T^2 \left[\frac{z^2 + z}{(z-1)^3} \right]_{z \rightarrow ze^{-aT}} \\
 &= T^2 \left[\frac{z^2 e^{2bT} + ze^{bT}}{(ze^{bT}-1)^3} \right] \\
 &= T^2 e^{-bT} \left[\frac{z(z + e^{-bT})}{(z - e^{-bT})^3} \right]
 \end{aligned}$$

(ix) e^{4ixn}

Solution: Since we have,

$$Z\{e^{at} f(t)\} = (Z\{f(t)\})_{z \rightarrow ze^{-aT}} \quad \text{for } t = nT \quad \text{and} \quad Z(1) = \frac{z}{z-1}$$

Here,

$$\begin{aligned}
 Z\{e^{4ixn}\} &= (Z(1))_{z \rightarrow ze^{-4i\pi}} \\
 &= \left(\frac{z}{z-1} \right)_{z \rightarrow ze^{-4i\pi}} \\
 &= \frac{ze^{-4i\pi}}{ze^{-4i\pi}-1} = \frac{z}{z - e^{4i\pi}} \\
 &= \frac{z}{z - \cos 4\pi - i \sin 4\pi} \times \frac{z - \cos 4\pi + i \sin 4\pi}{z - \cos 4\pi + i \sin 4\pi} \\
 &= \frac{z(z - \cos 4\pi) + iz \sin 4\pi}{(z - \cos 4\pi)^2 + \sin^2 4\pi} \\
 &= \frac{z(z - \cos 4\pi) + iz \sin 4\pi}{z^2 - 2z \cos 4\pi + 1}
 \end{aligned}$$

$$\Rightarrow Z(\cos 4\pi n + i \sin 4\pi n) = \frac{z(z - \cos 4\pi) + iz \sin 4\pi}{z^2 - 2z \cos 4\pi + 1}$$

Comparing the real part,

$$Z(\cos 4\pi n) = \frac{z(z - \cos 4\pi)}{z^2 - 2z \cos 4\pi + 1}$$

(x) $\cos\left(\frac{n\pi}{3}\right)$

Solution: Here,

$$\cos\left(\frac{n\pi}{3} + 5\right) = \cos\left(\frac{n\pi}{3}\right) \cos 5 - \sin\left(\frac{n\pi}{3}\right) \sin 5 \quad \dots (i)$$

Since we have,

$$Z\{e^{at} f(t)\} = (Z\{f(t)\})_{z \rightarrow ze^{-aT}} \quad \text{for } t = nT \quad \text{and} \quad Z(1) = \frac{z}{z-1}$$

3. **Unit Heaviside function**

A function $f(x)$ is said to be unit Heaviside function if it satisfies.

$$f_a(x) = f(x-a) = \begin{cases} 1 & \text{for } x < a \\ 0 & \text{for } x > a \end{cases} \quad \text{with } x \geq 0.$$

4. **Polynomial function:**

A function $f(x)$ is called polynomial function if it defined as

$$f(x) = \begin{cases} a^x & \text{for } x = 0, 1, 2, \dots \\ 0 & \text{for } x < 0 \end{cases}$$

where, a is a constant.

5. **Exponential function:**

An exponential function $f(x)$ is defined as,

$$f(x) = \begin{cases} e^{ax} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

where, a be a constant.

6. **Sinusoidal function:**

A sinusoidal function $f(x)$ is defined as,

$$f(x) = \begin{cases} \sin wx & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

7. **Unit impulse or Direc-delta function:**

A function $f(x)$ is called unit impulse function, is defined as,

$$f(x) = \begin{cases} 1 & \text{for } x = 0 \\ 0 & \text{for } x \neq 0 \end{cases}$$

Linearity property of Z-Transform

Statement: If $f(t)$ and $g(t)$ are any two function of t in the discrete time period $0, T, 2T, \dots$, then, $Z[af(t) + bg(t)] = aZ[f(t)] + bZ[g(t)]$, where a and b are two constants.

Proof: Let $f(t)$ and $g(t)$ are any two function of t in the discrete time period $0, T, 2T, \dots$, then,

$$\begin{aligned} Z[af(t) + bg(t)] &= az[f(t)] + bz[g(t)] z^{-n} \\ &= a \sum_{n=0}^{\infty} f(nT) z^{-n} + b \sum_{n=0}^{\infty} g(nT) z^{-n} \\ &= aZ[f(t)] + bZ[g(t)] \end{aligned}$$

This proves that Z-Transform is linear.

First Shifting Theorem

Statement: If $Z[f(t)] = F(z)$ then $z[e^{-at} f(t)] = F(e^{aT}z) = [f(z)]_{z \rightarrow ze^{aT}}$

Proof: Let,

$$Z[f(t)] = F(z) = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

Now,

$$Z[e^{-at} f(t)] = \sum_{n=0}^{\infty} e^{-anT} f(nT) z^{-n}$$

$$= \sum_{n=0}^{\infty} e^{-anT} f(nT) z^{-n}$$

$$= \sum_{n=0}^{\infty} f(nT) (e^{aT}z)^{-n} = F[ze^{aT}] = [f(z)]_{z \rightarrow ze^{aT}}$$

$$\text{Thus, } Z[e^{-at} f(t)] = [f(z)]_{z \rightarrow ze^{aT}}.$$

Second Shifting Theorem:

Statement: If $Z[f(t)] = F(z)$ then $Z[f(t+T)] = z[F(z) - f(0)]$

Proof: Let

$$Z[f(t)] = F(z) = \sum_{n=0}^{\infty} f(nT) z^{-n} = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

Now,

$$\begin{aligned} Z[f(t+T)] &= \sum_{n=0}^{\infty} f(nT+T) z^{-n} \\ &= \sum_{n=0}^{\infty} f((n+1)T) z^{-n} \\ &= \sum_{n=0}^{\infty} f(nT+T) z^{-n} \end{aligned}$$

Put, $n+1 = k$ then,

$$\begin{aligned} &= \sum_{k=1}^{\infty} f(kT) z^{-(k-1)} \\ &= z \sum_{k=1}^{\infty} f(kT) z^{-k} \\ &= z \left[\sum_{k=0}^{\infty} f(kT) z^{-k} - f(0) \right] = z[F(z) - f(0)]. \end{aligned}$$

$$\text{Thus, } Z[f(t+T)] = z[F(z) - f(0)].$$

Theorem 1: Show that $Z\{t^k\} = -z T \frac{d}{dz} [Z\{t^{k-1}\}]$

Proof: We have,

$$Z\{t^k\} = \sum_{n=0}^{\infty} t^k z^{-n} = \sum_{n=0}^{\infty} (nT)^k z^{-n} \quad \text{and} \quad Z\{t^{k-1}\} = \sum_{n=0}^{\infty} (nT)^{k-1} z^{-n}$$

So,

$$\frac{d}{dz} [Z\{t^{k-1}\}] = \sum_{n=0}^{\infty} (nT)^{k-1} \frac{d}{dz} (z^{-n})$$

$$= \sum_{n=0}^{\infty} (-n) (nT)^{k-1} z^{-(n+1)}$$

$$= \left(\frac{-1}{zT} \right) \sum_{n=0}^{\infty} (nT)^k z^{-n} = \left(\frac{-1}{zT} \right) Z(t^k)$$

$$\text{Thus, } Z(t^k) = -zT \frac{d}{dz} [Z(t^{k-1})]$$

Theorem 2: Show that $Z(a^n f(t)) = F\left(\frac{z}{a}\right)$ where $Z(f(t)) = F(z)$.

Proof: We have,

$$Z(f(t)) = F(z) = \sum_{n=0}^{\infty} f(t) z^{-n} = \sum_{n=0}^{\infty} f(nT) z^{-n}.$$

Now,

$$Z(a^n f(t)) = \sum_{n=0}^{\infty} a^n f(t) z^{-n} = \sum_{n=0}^{\infty} f(t) \left(\frac{z}{a}\right)^{-n} = F\left(\frac{z}{a}\right).$$

Theorem 3: If $f(n) = 0$ for $n < 0$ and if $Z(f(t)) = F(z)$ then $Z(f(t-kT)) = z^{-k} F(z)$ for $n > 0, k < 0$.

Proof: Let, $Z(f(t)) = F(z)$.

Now,

$$Z(f(t-kT)) = \sum_{n=0}^{\infty} f(t-kT) z^{-n} = \sum_{n=0}^{\infty} f[(n-k)T] z^{-n}$$

Put, $n-k = m$ then,

$$Z(f(t-kT)) = \sum_{m=-k}^{\infty} f(mT) z^{-(m+k)}$$

$$= z^{-k} \sum_{m=0}^{\infty} f(mT) z^{-m} \quad [\because f(n) = 0, \text{ for all } n < 0]$$

$$= z^{-k} Z(f(t)).$$

$$\text{Thus, } Z(f(t-kT)) = z^{-k} Z(f(t)).$$

Theorem: If $f(t) = 0$ for $t < 0$ and $Z(f(t)) = F(z)$ for $t > 0$. Then

$$Z(f(t+kT)) = z^{-k} \left[F(z) - \sum_{n=0}^{k-1} f(nT) z^{-n} \right].$$

Proof: Let, $f(t) = 0$, for $t < 0$. And let $Z(f(t)) = F(z)$, for $t > 0$.

Now,

$$Z(f(t+kT)) = \sum_{n=0}^{\infty} f[(n+k)T] z^{-n}$$

Put, $n+k = m$ then,

$$= \sum_{m=k}^{\infty} f(mT) z^{-(m-k)}$$

$$= \sum_{m=k}^{\infty} f(mT) z^{-(m-k)} - \sum_{m=0}^{k-1} f(mT) z^{-(m-k)}$$

$$= z^{-k} \left[\sum_{m=k}^{\infty} f(mT) z^{-m} - \sum_{m=0}^{k-1} f(mT) z^{-m} \right]$$

$$= z^{-k} \left[F(z) - \sum_{n=0}^{k-1} f(nT) z^{-n} \right].$$

$$\text{Thus, } Z(f(t+kT)) = z^{-k} \left[F(z) - \sum_{n=0}^{k-1} f(nT) z^{-n} \right].$$

Initial value theorem:

If $Z(f(t)) = F(z)$ then, $f(0) = \lim_{z \rightarrow \infty} F(z)$ and $\lim_{z \rightarrow \infty} z F(z) = f(1)$.

$$\text{Proof: Let, } Z(f(t)) = F(z) = \sum_{n=0}^{\infty} f(t) z^{-n}$$

$$= f(0) + f(1) z^{-1} + f(2) z^{-2} + \dots$$

So,

$$\lim_{z \rightarrow \infty} F(z) = f(0) \quad [\because 1/z \rightarrow 0 \text{ as } z \rightarrow \infty]$$

Also, set $f(0) = 0$ then,

$$F(z) = f(1) z^{-1} + f(2) z^{-2} + \dots$$

So, $z F(z) = f(1) + f(2) z^{-1} + \dots$

Then, $\lim_{z \rightarrow \infty} z F(z) = f(1)$.

Final value theorem:

If $Z(f(t)) = F(z)$ then, $\lim_{t \rightarrow \infty} f(t) = \lim_{z \rightarrow 1} (z-1) F(z)$

Proof: Let, $Z(f(t)) = F(z)$ and we have $Z(f(t+T)) = F(z) - z f(0)$

Now,

$$\lim_{z \rightarrow 1} [z F(z) - z f(0) - F(z)]$$

$$= \lim_{z \rightarrow 1} [Z(f(t+T)) - F(z)]$$

$$= \lim_{z \rightarrow 1} \left(\sum_{n=0}^{\infty} f(t+T) z^{-n} - F(z) \right)$$

$$= \lim_{z \rightarrow 1} \sum_{n=0}^{\infty} [f(nT+T) - f(nT)] z^{-n}$$

2007 Spring Q. No. 4(b) OR

Find the solution of one dimensional wave equation with initial deflection $\frac{1}{2} \sin 3x + \sin x$ and initial velocity is zero.

2011 Fall Q. No. 2(b)

Find the solution of one dimensional wave equation corresponding to the triangular initial deflection: $f(x) = \frac{2K}{L}x$, if $0 < x < L/2$, $= \frac{2K}{L}(L-x)$, if $L/2 < x < L$; with its initial velocity zero and $c = 1$.

Problems related to one dimensional heat equation**Theoretical Problems:****2003 Fall Q. No. 3(a)**

Writing one-dimensional heat equation with corresponding initial and boundary conditions solve it by using separation of variables.

OR 2011 Spring Q. No. 6(a)

Write one-dimensional heat equation and solve it completely.

2005 Spring Q. No. 4(a)

Find the solution of one dimensional heat equation of a rod of length L . With boundary condition $u(0, t) = 0 = u(L, t)$ for all t and initial condition $u(x, 0) = f(x)$.

OR 2007 Spring Q. No. 4(a)

Find the solution of one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$, having zero temperature in end points and initial temperature $f(x)$.

2008 Spring Q. No. 4(a); 2011 Fall Q. No. 3(b)

Derive one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ with required assumption.

PROBLEMS**2002 Q. No. 4(a)**

A rod of length l with insulated sides is initially at a uniform temperature u_0 . Its ends are suddenly cooled to 0°C and are kept at that temperature. Find the temperature function $u(x, t)$.

2004 Spring Q. No. 4(a)

Find the temperature $u(x, t)$ in a laterally insulated copper bar 80 cm long if the initial temperature is $100 \sin\left(\frac{\pi x}{80}\right)^\circ\text{C}$ and the ends are kept at 0°C . How long will it take for the maximum temperature in the bar to drop to 50°C ?

Physical data for Copper:

Density: 8.92 gm/cm^3

Specific heat: $0.092 \text{ cal/gm}^\circ\text{C}$

Thermal conductivity: $0.95 \text{ cal/cm sec}^\circ\text{C}$

Problems related to two dimensional equation**Theoretical Problems:****2003 Fall Q. No. 3(b)**

Write the Laplacian in spherical coordination. Also write the potential function with boundary and condition at infinity and obtain Legendre equation from it.

2004 Fall Q. No. 4(b)

Solve the equation: $\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right)$ where $u(r, t)$ represents the vibration of the circular membrane in time t , under the conditions

$$(i) \quad u(R, t) = 0 \text{ for all } t \geq 0 \quad (ii) \quad u(r, 0) = f(r) \quad (iii) \quad \left. \frac{\partial u}{\partial t} \right|_{t=0} = g(r)$$

2005 Spring Q. No. 5(a) OR

Define potential function and then find the solution of potential function by spherical membrane.

2005 Fall Q. No. 4(b)

Derive two dimensional heat equation, and solve $u_x - u_y = 0$ by using separation of variables.

2005 Fall Q. No. 5(a); 2007 Spring Q. No. 5(a)

What is Helmholtz's equation on $F(x, y)$ and solve it subject to $F(0, y) = 0 = F(a, y) = F(x, 0) = F(x, b)$.

OR

Derive Helmholtz equation and then solve it using initial and boundary equation.

2005 Fall Q. No. 5(a) OR

Write down Laplacian in two dimensions. Express Laplacian polar coordinate. Write down Laplacian in cylindrical co-ordinate.

OR 2005 Spring Q. No. 5(a)

Write down Laplacian in polar co-ordinate and cylindrical ordinates.

2006 Fall Q. No. 3(b); 2011 Spring 6(a) OR

Define potential function and find its solution by spherical membrane.

2008 Spring Q. No. 5(a)

Derive two dimensional heat equation and wave equation with required assumptions.

2008 Spring Q. No. 5(a) OR

Solve two dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$, in circular membrane.

2008 Fall Q. No. 3(a) OR

Show that the Laplacian in cylindrical polar coordinate is $u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} + u_{zz}$

2009 Fall Q. No. 3(b) OR

Show that the Laplacian in u in polar coordinate is $\nabla^2 u = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta}$

2009 Spring Q. No. 5(a)

Find the solution of two dimensional wave equation in circular membrane with necessary initial and boundary condition.

2012 Fall Q. No. 1(b)

Write the wave equation for vibrating circular membrane together with its initial and boundary conditions and solve it.

Separation

SHORT QUESTIONS**2004 Fall Q. No. 7(c)**

Write down the one-dimensional wave equation stating boundary and initial conditions. What is the Laplacian of u ?

2004 Spring 7(e)

What is Laplacian of u ? Write down the Laplacian of u in polar and cylindrical coordinates.

2008 Fall Q. No. 7(d); 2009 Fall Q. No. 7(d)

State the two dimensional wave equations in polar form.

2007 Spring Q. No. 7(a)

Solve $U_{xy} + U_x = 0$.

2007 Fall Q. No. 7(c)

Solve $u_x + u_y = 0$, by using separation of variables.

2008 Spring Q. No. 7(e); 2011 Spring Q. No. 7(c)

Solve the partial differential equation $u_{xx} - u_{yy} = 0$, by using separation of variables methods.

□ □ □

Unit 8**Z TRANSFORM****Definition (Z-Transform)**

Let $f(t)$ be the function of time t defined in the discrete time invariant system $0, T, 2T, \dots$, then the z -transform of $f(t)$ is defined by

$$Z\{f(t)\} = \sum_{n=0}^{\infty} f(nT)z^{-n},$$

where z be the complex number in the region where $\left|\frac{1}{z}\right| < R$ and $t = nT$.

Maclaurin's series and Binomial expansion gives

$$1. e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$2. \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$$

$$3. \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$4. \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$5. \cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$6. \sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

$$7. \log(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \text{ for } |x| < 1$$

$$8. \log(1-x) = -\sum_{n=0}^{\infty} \frac{x^n}{n}$$

$$9. \frac{a}{1-r} = a + ar + ar^2 + \dots \text{ for } |r| < 1 \text{ and } a \neq 0.$$

$$10. (1+x)^n = 1 + \frac{n}{1!}x + \frac{n(n-1)}{2!}x^2 + \dots + \frac{n(n-1)(n-2+1)}{r!}x^r + \dots \text{ for } |x| < 1$$

$$11. \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \dots$$

$$12. \frac{1}{(1+x)^2} = 1 - 2x + 3x^2 - \dots$$

Some functions:**1. Unit step function:**

A function $f(x)$ is called unit step function if it satisfies,

$$f(x) = \begin{cases} 1 & \text{for } x = 0, 1, 2, 3, \dots \\ 0 & \text{for } x < 0 \end{cases}$$

2. Unit Ramp function:

A function $f(x)$ is called unit Ramp function if it satisfies the condition,

$$f(x) = \begin{cases} x & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

$$B_{nm} = 4k \left(-\frac{2 \cos m\pi}{m^3 \pi^3} + \frac{2}{m^3 \pi^3} \right) \left(-\frac{2 \cos n\pi}{n^3 \pi^3} + \frac{2}{n^3 \pi^3} \right)$$

$$= \frac{16k}{m^3 n^3 \pi^6} (1 - \cos m\pi) (1 - \cos n\pi)$$

Again,

$$1 - \cos m\pi = 0, \text{ when } m \text{ is even}$$

$$1 - \cos n\pi = 0, \text{ when } n \text{ is even}$$

Now, when m and n are odd

$$1 - \cos m\pi = 1 - (-1) = 2$$

$$1 - \cos n\pi = 1 - (-1) = 2$$

$$\text{Hence, } B_{nm} = \frac{16k}{m^3 n^3 \pi^6} \cdot 2 \cdot 2 = \frac{64k}{m^3 n^3 \pi^6}$$

$$\text{and, } \lambda_{nm} = c\pi \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}} = \pi \sqrt{m^2 + n^2} = \pi \sqrt{m^2 + n^2}$$

The required solution is

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{64k}{m^3 n^3 \pi^6} \cos(\sqrt{m^2 + n^2} \cdot t) \sin m\pi x \sin n\pi y,$$

for m and n are odd.

Thus,

$$u(x, y, t) = \frac{64k}{\pi^6} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{m^3 n^3} \cos(\pi \sqrt{m^2 + n^2} \cdot t) \sin m\pi x \sin n\pi y$$

for odd value of m and n .

2. Find $u(x, y, t)$ for the rectangular membrane with sides a and b with $c = 1$, if the initial velocity is zero and initial deflection is $\sin \frac{2\pi x}{a} \sin \frac{3\pi y}{b}$.

Solution: Given initial deflection $f(x, y) = \sin \frac{2\pi x}{a} \sin \frac{3\pi y}{b}$ initial velocity, $g(x, y) = 0$.

The required solution is

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{nm} \cos \lambda_{nm} t \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$$

$$\text{Since } B_{nm}^* = 0 \quad [\because g(x, y) = 0]$$

Here,

$$B_{nm} = \frac{4}{ab} \int_0^a \int_0^b f(x, y) \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy$$

$$= \frac{4}{ab} \int_0^b \int_0^a \sin \frac{2\pi x}{a} \sin \frac{3\pi y}{b} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} dx dy$$

$$= \frac{1}{ab} \left[\int_0^a 2 \sin \frac{2\pi x}{a} \sin \frac{m\pi x}{a} dx \right] \left[\int_0^b 2 \sin \frac{3\pi y}{b} \sin \frac{n\pi y}{b} dy \right]$$

$$= \frac{1}{ab} \int_0^a \left[\cos \frac{(2-m)\pi x}{a} - \cos \frac{(2+m)\pi x}{a} \right] dx$$

$$\int_0^b \left[\cos \frac{(3-n)\pi y}{b} - \cos \frac{(3+n)\pi y}{b} \right] dy$$

$$= \frac{1}{ab} \left\{ \frac{\sin \left(\frac{2-m}{a} \right) \pi x}{\left(\frac{2-m}{a} \right) \pi} - \frac{\sin \left(\frac{2+m}{a} \right) \pi x}{\left(\frac{2+m}{a} \right) \pi} \right\}_0^a$$

$$\left\{ \frac{\sin \left(\frac{3-n}{b} \right) \pi y}{\left(\frac{3-n}{b} \right) \pi} - \frac{\sin \left(\frac{3+n}{b} \right) \pi y}{\left(\frac{3+n}{b} \right) \pi} \right\}_0^b$$

On further calculation, we get

$$B_{23} = 1 \quad \text{and}$$

$$\lambda_{23} = c\pi \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}} = \pi \sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}} = \pi \sqrt{\frac{4}{a^2} + \frac{9}{b^2}}$$

Thus the required solution is

$$u(x, y, t) = \cos \left[\pi \sqrt{\frac{4}{a^2} + \frac{9}{b^2}} t \right] \sin \left(\frac{2\pi x}{a} \right) \sin \left(\frac{3\pi y}{b} \right)$$

3. Find the vibration of a rectangular membrane of sides $a = 4\text{ ft}$ and $b = 2\text{ ft}$, if the tension is 12.5 lb/ft , the density is 2.5 slugs/ft^2 , the initial velocity is 0 and the initial displacement is $f(x, y) = 0.1 (4x - x^2) (2y - y^2) \text{ ft}$

[2007 Spring Q. No. 5(a) OR]

Solution: Given that,

$$f(x, y) = 0.1 (4x - x^2) (2y - y^2) \text{ ft} \quad \text{and} \quad g(x, y) = 0$$

$$\text{Also, } c^2 = \frac{T}{\rho} = \frac{12.5}{2.5} = 5$$

$$\text{Then, } B_{nm}^* = 0 \text{ since } g(x, y) = 0$$

Here,

$$B_{mn} = \frac{4}{4.2} \int_0^2 \int_0^4 0.1 (4x - x^2) (2y - y^2) \sin \frac{m\pi x}{4} \cdot \sin \frac{n\pi y}{2} dx dy$$

$$= \frac{1}{20} \int_0^4 (4x - x^2) \sin \frac{m\pi x}{4} dx \int_0^2 (2y - y^2) \sin \frac{n\pi y}{2} dy$$

Integrating first integral we get,

$$= \frac{128}{m^3 \pi^3} [1 - (-1)^m]$$

$$= \frac{256}{m^3 \pi^3} \quad (\text{for } m \text{ odd})$$

and for the second integral

$$= \frac{16}{n^3 \pi^3} [1 - (-1)^n]$$

$$= \frac{32}{n^3 \pi^3} \quad (\text{for } n \text{ odd})$$

for even m or n we get zero value of the integral.

Now,

$$B_{mn} = \frac{256 \times 32}{20 m^3 n^3 \pi^6} \approx \frac{0.426050}{m^3 n^3} \quad (\text{for } m \text{ \& } n \text{ both odd})$$

Thus, the solution of the problem is

$$u(x, y, t) = 0.426050 \sum_{m, n \text{ odd}} \sum \frac{1}{m^3 n^3} \cos \left(\frac{5\pi}{4} \sqrt{m^2 + 4n^2} t \right) \sin \frac{m\pi x}{4} \cdot \sin \frac{n\pi y}{2}$$

OTHER IMPORTANT QUESTION FROM FINAL EXAM FOR PRACTICE

Problems related to one dimensional wave equation

Theoretical Problems:

2004 Spring Q. No. 4(b); 2011 Spring Q. No. 5(a); 2011 Fall Q. No. 3(a)

Solve one dimensional, wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ under the conditions

(i) $u(0, t) = 0$ (ii) $u(1, t) = 0$ (iii) $u(x, 0) = f(x)$ and (iv) $\left. \frac{\partial u}{\partial t} \right|_{t=0} = g(x)$

OR 2008 Fall Q. No. 3(a)

Show that the solutions of the one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ satisfying the boundary conditions $u(0, t) = 0$ and $u(L, t) = 0$ is

$$u_n(x, t) = (B_n \cos \lambda_n t + B_n^* \sin \lambda_n t) \sin \left(\frac{n\pi}{L} \right) x \quad (n = 1, 2, 3, \dots)$$

OR 2009 Spring Q. No. 4(a)

Find $u(x, t)$ from one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$.

2006 Spring Q. No. 4(a); 2007 Spring Q. No. 4(b); 2007 Fall Q. No. 4(a); 2009 Fall Q. No. 3(a); 2012 Fall Q. No. 4(a)

Derive one dimensional wave equation of an elastic string which is tightly stretched to the length L and two ends are fixed.

2006 Fall Q. No. 2(b)

Derive one-dimensional wave equation. Solve $u_x + u_y = 0$ by using separation of variables.

2007 Spring Q. No. 2(a) OR

Solve one-dimensional wave equation completely.

Problems

2002 Q. No. 4(b); 2008 Spring Q. No. 4(b)

Find $u(x, t)$ of the string of length $l = \pi$ when $c^2 = 1$, the initial velocity is zero

and the initial deflection is: $f(x) = \begin{cases} kx/a & \text{for } 0 < x < a \\ k(\pi - x)/\pi - a & \text{for } a < x < \pi \end{cases}$

2003 Fall Q. No. 2(b)

Find $u(x, t)$ of vibrating string of length $L = \pi$, initial velocity $g(x) = 0$, $c^2 = 1$ and initial deflection is $0.1x(\pi - x)$.

2004 Fall Q. No. 4(b) OR; 2005 Spring Q. No. 4(b); 2009 Fall Q. No. 3(b)

Solve one dimensional wave equation with initial deflection $\frac{1}{100} \sin 3x$ and initial velocity if zero with length $L = \pi$ and $c^2 = 1$.

OR

Find the deflection $u(x, t)$ of the string of length $L = \pi$ when $c^2 = 1$, the initial velocity is zero and the initial deflection is $0.01 \sin 3x$.

2005 Fall Q. No. 4(a)

Define partial differential equation. Write down one dimensional wave equation and heat equation and similarly two dimensional heat equation and wave equations. Solve one dimensional wave equation with initial deflection $0.01 \sin 3x$ and initial velocity zero and $L = \pi$, $c^2 = 1$.

Clearly $z^{n-1} F(z)$ has simple poles at $z = -1, -2$.

Here, residue of $z^{n-1} F(z)$ at $z = -1$ is,

$$\begin{aligned}\text{Res}_{z=-1} z^{n-1} F(z) &= \lim_{z \rightarrow -1} (z+1) z^{n-1} F(z) \\ &= \lim_{z \rightarrow -1} \left(\frac{3z^{n+1} + 2z^n + z^{n-1}}{z+2} \right) \\ &= 3(-1)^{n+1} + 2(-1)^n + (-1)^{n-1} \\ &= (-1)^{n-1} (3 - 2 + 1) \\ &= 2(-1)^{n-1}\end{aligned}$$

And, residue of $z^{n-1} F(z)$ at $z = -2$ is,

$$\begin{aligned}\text{Res}_{z=-2} z^{n-1} F(z) &= \lim_{z \rightarrow -2} \left(\frac{3z^{n+1} + 2z^n + z^{n-1}}{z+1} \right) \\ &= \frac{3(-2)^{n+1} + 2(-2)^n + (-2)^{n-1}}{-1} \\ &= -(-2)^{n-1} (12 - 4 + 1) \\ &= -9(-2)^{n-1}\end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = -9(-2)^{n-1} + 2(-1)^{n-1}$$

$$(xviii) \frac{z(1 - e^{-at})}{(z-1)(z - e^{-at})}$$

Solution: Let,

$$F(z) = \frac{z(1 - e^{-at})}{(z-1)(z - e^{-at})}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^n (1 - e^{-at})}{(z-1)(z - e^{-at})}$$

Clearly, $z^{n-1} F(z)$ has simple poles at $z = 1$ and e^{-at} .

Here, residue of $z^{n-1} F(z)$ at $z = 1$ is,

Now,

$$\text{Res}_{z=1} z^{n-1} F(z) = \lim_{z \rightarrow 1} (z-1) z^{n-1} F(z) = \lim_{z \rightarrow 1} \frac{z^n (1 - e^{-at})}{z - e^{-at}} = 1$$

And, residue of $z^{n-1} F(z)$ at $z = e^{-at}$ is,

$$\text{Res}_{z=e^{-at}} z^{n-1} F(z) = \lim_{z \rightarrow e^{-at}} \frac{z^n (1 - e^{-at})}{z-1} = -e^{-atn}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = 1 - e^{-atn}$$

$$(xix) \frac{z^2}{(1-z^2)^3}$$

Solution: Let,

$$F(z) = \frac{z^2}{(1-z^2)^3}$$

Then,

$$z^{n-1} F(z) = \frac{z^{n+2}}{(1-z^2)^3} = \frac{z^{n+2}}{z^3(z-1)^3} = \frac{z^n}{(z-1)^3}$$

Clearly, $z^{n-1} F(z)$ has pole at $z = 1$ of degree 3.

Here, residue of $z^{n-1} F(z)$ at $z = 1$ is,

residue of $z^{n-1} F(z)$ at $z = p$ of order r is,

$$\text{Residue} = \lim_{z \rightarrow 2} \frac{1}{(r-1)!} \frac{d^{r-1}}{dz^{r-1}} [(z-p)^r F(z)]$$

$$\begin{aligned}\text{Res}_{z=1} z^{n-1} F(z) &= \lim_{z \rightarrow 1} \frac{1}{(3-1)!} \frac{d^2}{dz^2} [z^n] \\ &= \frac{1}{2!} \lim_{z \rightarrow 1} \frac{d^2}{dz^2} (z^n) \\ &= \frac{1}{2} n(n-1) \lim_{z \rightarrow 1} (z^{n-2}) = \frac{1}{2} (n^2 - n)\end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = \frac{1}{2} (n^2 - n)$$

$$(xx) \frac{z^{-1}(1-z^2)}{(1+z^2)^2}$$

Solution: Let,

$$\begin{aligned}F(z) &= \frac{z^{-1}(1-z^2)}{(1+z^2)^2} = \frac{1}{2} \left(1 - \frac{1}{z^2}\right) \cdot \left(1 + \frac{1}{z^2}\right)^{-2} \\ &= \frac{1}{2} (z^2 - 1) \cdot \frac{1}{z^2 - 4} (z^2 + 1)^{-2} \\ &= \frac{z(z^2 - 1)}{(z^2 + 1)^2} \\ &= \frac{z^3 - z}{(z-i)^2 (z+i)^2}\end{aligned}$$

Clearly, $F(z)$ has poles at $z = i$ of order 2 and at $z = -i$ of order 2.

Here, residue of $z^{n-1} F(z)$ at $z = i$ of order 2 is,

$$\begin{aligned}\text{Res}_{z=i} z^{n-1} F(z) &= \lim_{z \rightarrow i} \frac{1}{(2-1)!} \frac{d^{2-1}}{dz^{2-1}} [z^{n-1} F(z) (z-i)^2] \\ &= \lim_{z \rightarrow i} \frac{d}{dz} \left(\frac{z^{n+2} - z^n}{(z+i)^2} \right) \\ &= \lim_{z \rightarrow i} \frac{(z+i)^2 [(n+2)z^{n+1} - nz^{n-1}] - (z^{n+2} - z^n) 2(z+i)}{(z+i)^4} \\ &= \frac{(2i)^2 [(n+2)i^{n+1} - ni^{n-1}] - (i^{n+2} - i^n) 2 \cdot 2i}{(2i)^4}\end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = \frac{11}{6}(4)^n + \frac{1}{6}(-2)^n$$

$$(xiv) \frac{2z^2 - 5z}{(z-2)(z-3)}$$

Solution: Let,

$$F(z) = \frac{2z^2 - 5z}{(z-2)(z-3)}$$

$$\text{Then, } z^{n-1} F(z) = \frac{2z^{n+1} - 5z^n}{(z-2)(z-3)}$$

Clearly $z^{n-1} F(z)$ has simple poles at $z=2$ and 3 .

Here, residue of $z^{n-1} F(z)$ at $z=2$ is,

$$\begin{aligned} \text{Res}_{z=2} z^{n-1} F(z) &= \lim_{z \rightarrow 2} (z-2) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 2} \frac{2z^{n+1} - 5z^n}{(z-3)} = \frac{2 \cdot 2^{n+1} - 5 \cdot 2^n}{-1} = \frac{2^n(4-5)}{-1} = 2^n \end{aligned}$$

And, residue of $z^{n-1} F(z)$ at $z=3$ is,

$$\begin{aligned} \text{Res}_{z=3} z^{n-1} F(z) &= \lim_{z \rightarrow 3} (z-3) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 3} \frac{2z^{n+1} - 5z^n}{(z-2)} = \frac{2(3)^{n+1} - 5 \cdot 3^n}{1} = \frac{3^n(6-5)}{1} = 3^n \end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = 2^n + 3^n.$$

$$(xv) \frac{2z^3 + z}{(z-2)^2(z-1)}$$

Solution: Here,

$$F(z) = \frac{2z^3 + z}{(z-2)^2(z-1)}$$

$$\text{Then, } z^{n-1} F(z) = \frac{2z^{n+2} + z^n}{(z-2)^2(z-1)}$$

Clearly, $z^{n-1} F(z)$ has simple poles at $z=1$ and at $z=2$ of degree 2.

Here, residue of $z^{n-1} F(z)$ at $z=1$ is,

$$\begin{aligned} \text{Res}_{z=1} z^{n-1} F(z) &= \lim_{z \rightarrow 1} (z-1) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 1} (z-1) \left(\frac{2z^{n+2} + z^n}{(z-1)(z-2)^2} \right) = \frac{2+1}{(1-2)^2} = 3. \end{aligned}$$

And, residue of $z^{n-1} F(z)$ at $z=2$ is,

$$\text{Res}_{z=2} z^{n-1} F(z) = \lim_{z \rightarrow 2} (z-2) z^{n-1} F(z)$$

$$= \lim_{z \rightarrow 2} \frac{d}{dz} [(z-2)^2 z^{n-1} F(z)]$$

$$= \lim_{z \rightarrow 2} \frac{d}{dz} \left(\frac{2z^{n+2} + z^n}{z-1} \right)$$

$$= \lim_{z \rightarrow 2} \left[\frac{(2(n+2)z^{n+1} + nz^{n-1})(z-1) - (2z^{n+2} + z^n)}{(z-1)^2} \right]$$

$$= 2(n+2)2^{n+1} + n(2)^{n-1} - 2(2)^{n+2} - 2^n$$

$$= 2^{n-1}(8n+16+n) - 8n^n - 2^n$$

$$= 9n(2^{n-1}) + 2^4(2^{n-1}) - 9(2^n)$$

$$= 9n(2^{n-1}) + 2^n(8-9)$$

$$= 9n(2)^{n-1} - 2^n$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = 9n(2)^{n-1} - 2^n + 3$$

$$(xvi) \frac{z}{(z-1)^2(z-2)}$$

Solution: Let,

$$F(z) = \frac{z}{(z-1)^2(z-2)}$$

Then,

$$z^{n-1} F(z) = \frac{z^n}{(z-1)^2(z-2)}$$

Clearly $z^{n-1} F(z)$ has simple pole at $z=2$ and $z=1$ of degree 2.

Here, residue of $z^{n-1} F(z)$ at $z=2$ is,

$$\text{Res}_{z=2} z^{n-1} F(z) = \lim_{z \rightarrow 2} ((z-2)^2 z^{n-1} F(z)) = \lim_{z \rightarrow 2} \frac{z^n}{(z-1)^2} = 2^n$$

And, residue of $z^{n-1} F(z)$ at $z=1$ is,

$$\begin{aligned} \text{Res}_{z=1} z^{n-1} F(z) &= \lim_{z \rightarrow 1} \frac{d}{dz} ((z-1)^2 z^{n-1} F(z)) \\ &= \lim_{z \rightarrow 1} \frac{d}{dz} \left(\frac{z^n}{z-2} \right) = \lim_{z \rightarrow 1} \frac{nz^{n-1}(z-2) - z^n}{(z-2)^2} = -n-1 \end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = 2^n - n - 1$$

$$(xvii) \frac{3z^2 + 2z + 1}{z^2 + 3z + 2}$$

Solution: Let,

$$F(z) = \frac{3z^2 + 2z + 1}{z^2 + 3z + 2}$$

$$\text{Then, } z^{n-1} F(z) = \frac{3z^{n+1} + 2z^n + z^{n-1}}{(z+1)(z+2)}$$

$$\begin{aligned}\lim_{z \rightarrow 1} z^{n-1} F(z) &= \lim_{z \rightarrow 1} \frac{d}{dz} \left((z-1)^2 \frac{z^{n+1} + 2z^n}{(z-1)^2} \right) \\ &= \lim_{z \rightarrow 1} [(n+1)z^2 + 2nz^{n-1}] = 3n+1\end{aligned}$$

We know,

$$Z^{-1} \{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1} \{F(z)\} = 3n+1$$

$$(x) \frac{z+2}{(z-2)z^2}$$

$$\text{Solution: Let, } F(z) = \frac{z+2}{(z-2)z^2}$$

$$\text{Then } z^{n-1} F(z) = \frac{z^n + 2z^{n-1}}{(z-2)z^2}$$

Clearly $z^{n-1} F(z)$ has at $z=2$ of simple pole and at $z=0$ of order of 2.

Here the residue of $z^{n-1} F(z)$ at $z=2$ is

$$\begin{aligned}\text{Res}_{z=2} z^{n-1} F(z) &= \lim_{z \rightarrow 2} \frac{z^n + 2z^{n-1}}{z^2} \\ &= \frac{2^n + 2^{n-1} \cdot 2}{2^2} = \frac{2 \cdot 2^n}{2^2} = 2^{n-1}\end{aligned}$$

And, the residue of $z^{n-1} F(z)$ at $z=0$ is

$$\begin{aligned}\text{Res}_{z=0} z^{n-1} F(z) &= \lim_{z \rightarrow 0} \frac{d}{dz} \left(\frac{z^n + 2z^{n-1}}{z-2} \right) \\ &= \lim_{z \rightarrow 0} \left(\frac{(nz^{n-1} + 2(n-1)z^{n-2})(z-2) - (z^n + 2z^{n-1})}{(z-2)^2} \right) \\ &= 0\end{aligned}$$

Since we know,

$$Z^{-1} \{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1} \{F(z)\} = 2^{n-1} + 0 = 2^{n-1}$$

$$(xi) \frac{z}{(z+1)^2}$$

$$\text{Solution: Let } F(z) = \frac{z}{(z+1)^2}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^n}{(z+1)^2}$$

Clearly $z^{n-1} F(z)$ has pole at $z=-1$ of order 2.

Then, the residue of $z^{n-1} F(z)$ at $z=-1$ be,

$$\begin{aligned}\text{Res}_{z=-1} z^{n-1} F(z) &= \lim_{z \rightarrow -1} \frac{1}{(z-1)!} \frac{d^{2-1}}{dz^{2-1}} [z^{n-1} F(z) (z+1)^2] \\ &= \lim_{z \rightarrow -1} \frac{d}{dz} (z^n)\end{aligned}$$

$$= \lim_{z \rightarrow -1} nz^{n-1} = n(-1)^{n-1}$$

We know,

$$Z^{-1} \{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1} \{F(z)\} = n(-1)^{n-1}.$$

$$(xii) \frac{1}{(z-2)(z-3)}$$

$$\text{Solution: Let, } F(z) = \frac{1}{(z-2)(z-3)}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^{n-1}}{(z-2)(z-3)}$$

Clearly $z^{n-1} F(z)$ has simple poles at $z=2$ and $z=3$.

Here the residue of $z^{n-1} F(z)$ at $z=2$ is

$$\text{Res}_{z=2} z^{n-1} F(z) = \lim_{z \rightarrow 2} (z-2) z^{n-1} F(z) = \lim_{z \rightarrow 2} \frac{z^{n-1}}{z-3} = - (2)^{n-1}$$

And, the residue of $z^{n-1} F(z)$ at $z=3$ is

$$\text{Res}_{z=3} z^{n-1} F(z) = \lim_{z \rightarrow 3} (z-3) z^{n-1} F(z) = \lim_{z \rightarrow 3} \frac{z^{n-1}}{z-2} = (3)^{n-1}$$

We know,

$$Z^{-1} \{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1} \{F(z)\} = (3)^{n-1} - (2)^{n-1}$$

$$(xiii) \frac{2z^2 + 3z}{(z+2)(z-4)}$$

Solution: Let,

$$F(z) = \frac{2z^2 + 3z}{(z+2)(z-4)}$$

$$\text{Then, } z^{n-1} F(z) = \frac{2z^{n+1} + 3z^n}{(z+2)(z-4)}$$

Clearly $z^{n-1} F(z)$ has simple poles at $z=-2, 4$.

Here, residue of $z^{n-1} F(z)$ at $z=-2$ is,

$$\begin{aligned}\text{Res}_{z=-2} z^{n-1} F(z) &= \lim_{z \rightarrow -2} (z+2) z^{n-1} F(z) \\ &= \lim_{z \rightarrow -2} \frac{2z^{n+1} + 3z^n}{z-4} \\ &= \frac{2(-2)^{n+1} + 3(-2)^n}{-6} = \frac{(-2)^n (-4+3)}{-6} = \frac{1}{3} (-2)^{n-1} = \frac{1}{6} (-2)^n\end{aligned}$$

And, residue of $z^{n-1} F(z)$ at $z=4$ is,

$$\begin{aligned}\text{Res}_{z=4} z^{n-1} F(z) &= \lim_{z \rightarrow 4} (z-4) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 4} \frac{2z^{n+1} + 3z^n}{z+2} = \frac{2(4)^{n+1} + 3 \cdot 4^n}{6} = \frac{11}{6} (4)^n\end{aligned}$$

$$\text{So, } Z^{-1}\{F(z)\} = (1-n)2^n$$

$$(v) \frac{z^2 - z}{(z-2)(z-3)}$$

Solution: Let,

$$F(z) = \frac{z^2 - z}{(z-2)(z-3)}$$

Then,

$$z^{n-1} F(z) = \frac{z^{n+1} - z^n}{(z-2)(z-3)}$$

Clearly the form $z^{n-1} F(z)$ has poles at $z=2$ and $z=3$ of simple order.

Here, residue of $z^{n-1} F(z)$ at $z=2$ is,

$$\begin{aligned} \text{Res}_{z=2} z^{n-1} F(z) &= \lim_{z \rightarrow 2} (z-2) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 2} \frac{z^{n+1} - z^n}{z-3} = \frac{2^{n+1} - 2^n}{2-3} = \frac{2^n(2-1)}{-1} = -2^n \end{aligned}$$

And the residue of $z^{n-1} F(z)$ at $z=3$ is,

$$\begin{aligned} \text{Res}_{z=3} z^{n-1} F(z) &= \lim_{z \rightarrow 3} (z-3) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 3} \frac{z^{n+1} - z^n}{z-2} = \frac{3^{n+1} - 3^n}{3-2} = \frac{3^n(3-1)}{1} = 2(3^n) \end{aligned}$$

Since we have,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = 2(3^n) - 2^n$$

$$(vi) \frac{z}{z^2 - 3z + 2}$$

$$\text{Solution: Let, } F(z) = \frac{z}{z^2 - 3z + 2} = \frac{z}{(z-2)(z-1)}$$

Repeated question to Q.(iii)

$$(vii) \frac{10z + 5}{(z-1)(z-1/5)}$$

Solution: Let,

$$F(z) = \frac{10z + 5}{(z-1)(z-1/5)}$$

Then,

$$z^{n-1} F(z) = \frac{10z^n + 5z^{n-1}}{(z-1)(z-1/5)}$$

Clearly the function $z^{n-1} F(z)$ has poles at $z=1$ and at $z=1/5$ of single order.

Here, the residue of $z^{n-1} F(z)$ at $z=1$ is,

$$\text{Res}_{z=1} z^{n-1} F(z) = \lim_{z \rightarrow 1} (z-1) z^{n-1} F(z)$$

$$= \lim_{z \rightarrow 1} \frac{10z^n + 5z^{n-1}}{(z-1/5)} = \frac{10(1)^n + 5(1)^{n-1}}{1-1/5} = \frac{10+5}{4/5} = \frac{75}{4}$$

And the residue of $z^{n-1} F(z)$ at $z=1/5$ is,

$$\begin{aligned} \text{Res}_{z=1/5} z^{n-1} F(z) &= \lim_{z \rightarrow 1/5} (z-1/5) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 1/5} \left(\frac{10z^n + 5z^{n-1}}{z-1} \right) \\ &= \frac{10(1/5)^n + 5(1/5)^{n-1}}{1/5-1} \\ &= \frac{1}{5^{n-1}} \left[\frac{10(1/5) + 5}{-4/5} \right] = \frac{1}{5^{n-1}} \left(\frac{7}{-4} \right) = \frac{-7}{4(5^{n-1})} \end{aligned}$$

Since we have,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z).$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = \frac{1}{4} \left[75 - \frac{7}{5^{n-1}} \right]$$

$$(viii) \frac{z+1}{z^2 - 2z + 1}$$

Solution: Here,

$$F(z) = \frac{z+1}{z^2 - 2z + 1} = \frac{z+1}{(z-1)^2}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^n + z^{n-1}}{(z-1)^2}$$

Clearly $z^{n-1} F(z)$ has poles at $z=1$ of order 2.

Here, residue of $z^{n-1} F(z)$ at $z=1$ is,

$$\begin{aligned} \lim_{z \rightarrow 1} z^{n-1} F(z) &= \lim_{z \rightarrow 1} \frac{d}{dz} \left((z-1)^2 \frac{z^n + z^{n-1}}{(z-1)^2} \right) \\ &= \lim_{z \rightarrow 1} [nz^{n-1} + (n-1)z^{n-2}] = n+n-1 = 2n-1 \end{aligned}$$

Since we know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z).$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = 2n-1$$

$$(ix) \frac{z(z+2)}{(z-1)^2}$$

$$\text{Solution: Let, } F(z) = \frac{z(z+2)}{(z-1)^2}$$

$$\text{Then } z^{n-1} F(z) = \frac{z^{n+1} + 2z^n}{(z-1)^2}$$

Clearly $z^{n-1} F(z)$ has at $z=1$ of order of 2.

Here, the residue of $z^{n-1} F(z)$ at $z=1$ is,

$$= \lim_{z \rightarrow i} \frac{z^n}{(z-1)(z+i)} = \frac{i^n}{(i-1)(i+i)} = \frac{i^n}{(i-1)(2i)}$$

Also,

$$\begin{aligned} \text{Res}_{z=-i} z^{n-1} F(z) &= \lim_{z \rightarrow -i} (z+i) z^{n-1} F(z) \\ &= \lim_{z \rightarrow -i} \frac{z^n}{(z-1)(z-i)} = \frac{(-i)^n}{(-i-1)(-2i)} \end{aligned}$$

Now (ii) becomes,

$$\begin{aligned} Z^{-1}\{F(z)\} &= \frac{1}{2} + \frac{i^n}{(i-1)2i} + \frac{(-1)^n i^n}{2i(i+1)} \\ &= \frac{1}{2} + \frac{i^n(i+1) + (-i)^n(i-1)}{2i(i^2-1)} = \frac{1}{2} + i^{n+1} + i^n \end{aligned}$$

$$(ii) \frac{z^2 - 3z}{(z-5)(z+2)}$$

[2003 Fall Q. No. 4(b)]

Solution: Let

$$F(z) = \frac{z^2 - 3z}{(z-5)(z+2)}$$

Then,

$$z^{n-1} F(z) = \frac{z^{n+1} - 3z^n}{(z-5)(z+2)} \quad \dots\dots\dots(1)$$

Clearly $z^{n-1} F(z)$ has poles at $z=5$ and $z=-2$

Since we have,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z) \quad \dots\dots\dots(2)$$

Here, residue of $z^{n-1} F(z)$ at $z=5$ is

$$\begin{aligned} \text{Res}_{z=5} z^{n-1} F(z) &= \lim_{z \rightarrow 5} (z-5) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 5} \frac{z^n(z-3)}{z+2} = \frac{5^n(5-3)}{5+2} = \frac{2}{7}(5^n) \end{aligned}$$

And residue of $z^{n-1} F(z)$ at $z=-2$ is,

$$\begin{aligned} \text{Res}_{z=-2} z^{n-1} F(z) &= \lim_{z \rightarrow -2} (z+2) z^{n-1} F(z) \\ &= \lim_{z \rightarrow -2} \frac{z^n(z-3)}{z-5} = \frac{(-2)^n(-2-3)}{-2-5} = -\frac{5(-2)^n}{7} \end{aligned}$$

Now, (1) becomes,

$$Z^{-1}\{F(z)\} = \frac{2}{7}(5^n) - \left(\frac{5}{7}\right)(-2)^n$$

$$(iii) \frac{z}{(z-1)(z-2)}$$

Solution: Let,

$$F(z) = \frac{z}{(z-1)(z-2)} \quad \dots\dots\dots(1)$$

Then,

$$z^{n-1} F(z) = \frac{z^n}{(z-1)(z-2)} \quad \dots\dots\dots(2)$$

Since we have,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z) \quad \dots\dots\dots(3)$$

Clearly, (2) has poles at $z=1$ and $z=2$.Here, residue of (2) at $z=1$ is,

$$\begin{aligned} \text{Res}_{z=1} z^{n-1} F(z) &= \lim_{z \rightarrow 1} (z-1) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 1} \left(\frac{z^n}{z-2} \right) = \frac{1^n}{1-2} = -1^n \end{aligned}$$

And residue of (2) at $z=2$ is,

$$\begin{aligned} \text{Res}_{z=2} z^{n-1} F(z) &= \lim_{z \rightarrow 2} (z-2) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 2} \left(\frac{z^n}{z-1} \right) = \frac{2^n}{2-1} = 2^n \end{aligned}$$

Now (3) becomes,

$$\begin{aligned} Z^{-1}\{F(z)\} &= -1^n + 2^n \\ \Rightarrow Z^{-1}\{F(z)\} &= 2^n - 1^n \end{aligned}$$

$$(iv) \frac{z^2 - 4z}{(z-2)^2}$$

Solution: Let,

$$F(z) = \frac{z^2 - 4z}{(z-2)^2}$$

Then,

$$z^{n-1} F(z) = \frac{z^{n+1} - 4z^n}{(z-2)^2}$$

Clearly, $z^{n-1} F(z)$ pole at $z=2$ of order 2.So, residue of $z^{n-1} F(z)$ at $z=2$ is,

$$\begin{aligned} \text{Res}_{z=2} z^{n-1} F(z) &= \lim_{z \rightarrow 2} \frac{d}{dz} [(z-2)^2 z^{n-1} F(z)] \\ &= \lim_{z \rightarrow 2} \frac{d}{dz} (z^{n+1} - 4z^n) \\ &= \lim_{z \rightarrow 2} [(n+1)z^n - 4nz^{n-1}] \\ &= (n+1)2^n - 4n2^{n-1} \\ &= (n+1)2^n - 2n2^n \\ &= 2^n(n+1-2n) \\ &= (1-n)2^n \end{aligned}$$

Since we know that,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z) \text{ at } z=2 \text{ of all orders.}$$

$$= a^{-1} \left[\frac{a^3}{a^2} \left(\frac{z(z+a)}{(z-a)^2} \right) \right]$$

$$= \frac{z(z+a)}{(z-a)^2}$$

(xx) $[3(2)^n - 4(3)^n] u(n)$

Solution: Since we have,

$$Z\{a^n f(t)\} = (Z\{f(t)\})_{z \rightarrow az} \quad \text{and} \quad Z\{u(n)\} = \frac{z}{z-1}$$

Now,

$$\begin{aligned} Z\{[3(2)^n - 4(3)^n] u(n)\} &= 3[Z\{u(n)\}]_{z \rightarrow 2z} - 4[Z\{u(n)\}]_{z \rightarrow 3z} \\ &= 3 \left(\frac{z}{z-1} \right)_{z \rightarrow 2z} - 4 \left(\frac{z}{z-1} \right)_{z \rightarrow 3z} \\ &= 3 \left(\frac{z}{z-2} \right) - 4 \left(\frac{z}{z-3} \right) \\ &= \frac{3z(z-3) - 4z(z-2)}{(z-2)(z-3)} \\ &= \frac{z(3z-9-4z+8)}{(z-2)(z-3)} \\ &= \frac{-z(z+1)}{(z-2)(z-3)} \end{aligned}$$

(xxi) $\sin w_0 n \cdot u(n)$

Solution: Since we have,

$$Z\{e^{-at} f(t)\} = (Z\{f(t)\})_{z \rightarrow ze^{-aT}} \quad \text{and} \quad Z\{u(n)\} = \frac{z}{z-1}$$

Now,

$$\begin{aligned} Z\{\sin w_0 n \cdot u(n)\} &= Z\left\{ \frac{e^{jw_0 n} - e^{-jw_0 n}}{2j} u(n) \right\} \\ &= \frac{1}{2j} [(Z\{u(n)\})_{z \rightarrow ze^{-jw_0 T}} - (Z\{u(n)\})_{z \rightarrow ze^{jw_0 T}}] \\ &= \frac{1}{2j} \left[\left(\frac{z}{z-1} \right)_{z \rightarrow ze^{-jw_0 T}} - \left(\frac{z}{z-1} \right)_{z \rightarrow ze^{jw_0 T}} \right] \\ &= \frac{1}{2j} \left[\frac{z}{z - e^{-jw_0 T}} - \frac{z}{z - e^{jw_0 T}} \right] \\ &= \frac{1}{2j} \left[\frac{z(e^{jw_0 T} - e^{-jw_0 T})}{z^2 - z(e^{jw_0 T} + e^{-jw_0 T}) + 1} \right] \\ &= \frac{z \sin w_0}{z^2 - 2z \cos w_0 + 1} \end{aligned}$$

(xxii) $\delta(n-k)$ for $k > 0$.Solution: Since we have, $\delta(n) = \begin{cases} 1 & \text{for } n = 0 \\ 0 & \text{for } n \neq 0 \end{cases}$ Now, for $k > 0$,

$$Z\{\delta(n-k)\} = \sum_{n=0}^{\infty} \delta(n-k) z^{-n} = z^{-k} \quad \text{being } \delta(n-k) = \begin{cases} 1 & \text{for } n = k \\ 0 & \text{for } n \neq k \end{cases}$$

(xxiii) $(1+n) u(n)$

Solution: Since we have,

$$Z\{n f(t)\} = -z \frac{d}{dz} (Z\{f(t)\}) \quad \text{and} \quad Z\{u(n)\} = \frac{z}{z-1}$$

Now,

$$\begin{aligned} Z\{(1+n) u(n)\} &= Z\{u(n)\} + Z\{n u(n)\} \\ &= \frac{z}{z-1} + z \frac{d}{dz} \left(\frac{z}{z-1} \right) \\ &= \frac{z}{z-1} + z \frac{d}{dz} \left(\frac{z}{z-1} \right) \\ &= \frac{z}{z-1} + z \left(\frac{-1}{(z-1)^2} \right) \\ &= \frac{z(z-1) + z}{(z-1)^2} \\ &= \frac{z^2}{(z-1)^2} \end{aligned}$$

(xxiv) $[4(3)^n - (2)^n] u(n)$

Solution: Similar to (xx).

2. Find the inverse Z-transform of

$$(i) \frac{z}{(z-1)(z^2+1)}$$

Solution: Let,

$$f(z) = \frac{z}{(z-1)(z^2+1)} = \frac{z}{(z-1)(z-i)(z+i)}$$

Then,

$$z^{n-1} F(z) = \frac{z^n}{(z-1)(z-i)(z+i)} \quad \dots\dots\dots(i)$$

Clearly the poles of $z^{n-1} F(z)$ are $z = 1$, $z = i$ and $z = -i$.

Since we know that,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z) \quad \dots\dots\dots(ii)$$

Here, residue of $z^{n-1} F(z)$ at $z = 1$ is

$$\begin{aligned} \text{Res}_{z=1} z^{n-1} F(z) &= \lim_{z \rightarrow 1} (z-1) z^{n-1} F(z) \\ &= \lim_{z \rightarrow 1} \frac{z^n}{(z-i)(z+i)} = \lim_{z \rightarrow 1} \frac{z^n}{z^2+1} = \frac{1^n}{1+1} = \frac{1}{2} \end{aligned}$$

And,

$$\text{Res}_{z=i} z^{n-1} F(z) = \lim_{z \rightarrow i} (z-i) z^{n-1} F(z)$$

$$\begin{aligned}
&= -T^2 \left(z \frac{d}{dz} \right)^2 \left[z \left(\frac{-e^{-bT}}{(z - e^{-bT})^2} \right) \right] \\
&= -T^3 (-e^{-bT}) \left(z \frac{d}{dz} \right) \left[z \frac{d}{dz} \left(\frac{z}{(z - e^{-bT})^2} \right) \right] \\
&= T^3 e^{-bT} \left(z \frac{d}{dz} \right) \left[z \left(\frac{(z - e^{-bT})^2 - z \cdot 2(z - e^{-bT})}{(z - e^{-bT})^4} \right) \right] \\
&= T^3 e^{-bT} z \frac{d}{dz} \left[\frac{z^2 - ze^{-bT} - 2z^2}{(z - e^{-bT})^3} \right] \\
&= T^3 \frac{ze^{-bT} [(z - e^{-bT})^3 (2z - e^{-bT} - 4z) - (z^2 - e^{-bT}z - 2z^2) \cdot 3(z - e^{-bT})^2]}{(z - e^{-bT})^6} \\
&= T^3 \frac{ze^{-bT} [(z - e^{-bT}) (2z - e^{-bT} - 4z) - 3z^2 + 3e^{-bT}z + 6z^2]}{(z - e^{-bT})^4} \\
&= T^3 \frac{ze^{-bT} (2z^2 - ze^{-bT} - 4z^2 - 2ze^{-bT} + e^{-2bT} + 4ze^{-bT} - 3z^2 + 3e^{-bT}z + 6z^2)}{(z - e^{-bT})^4} \\
&= T^3 \frac{ze^{-bT} [z^2 + e^{-2bT} + 4ze^{-bT}]}{(z - e^{-bT})^4}
\end{aligned}$$

(xvi) $e^{-at} - e^{-bt}$

Solution: Since we have,

$$z(e^{-at}) = (z(1))_{z \rightarrow \infty} aT \quad \text{and} \quad Z(1) = \frac{z}{z-1}$$

Now,

$$\begin{aligned}
z(e^{-at} - e^{-bt}) &= Z(e^{-at}) - Z(e^{-bt}) \\
&= (z(1))_{z \rightarrow \infty} aT - (z(1))_{z \rightarrow \infty} bT \\
&= \left(\frac{z}{z-1} \right)_{z \rightarrow \infty} aT - \left(\frac{z}{z-1} \right)_{z \rightarrow \infty} bT \\
&= \frac{ze^{aT}}{ze^{aT} - 1} - \frac{ze^{bT}}{ze^{bT} - 1} \\
&= \frac{z^2 e^{aT} e^{bT} - ze^{aT} - z^2 e^{aT} e^{bT} + ze^{bT}}{(ze^{aT} - 1)(ze^{bT} - 1)} \\
&= \frac{z(e^{bT} - e^{aT})}{(ze^{aT} - 1)(ze^{bT} - 1)}
\end{aligned}$$

(xvii) $(1 - bt)e^{-bt}$

Solution: Since we have,

$$Z(e^{-bt} f(t)) = (Z\{f(t)\})_{z \rightarrow \infty} aT \quad \text{and} \quad Z\{t f(t)\} = -Tz \frac{d}{dz} (Z\{f(t)\}), \quad Z(1) = \frac{z}{z-1}$$

Now,

$$\begin{aligned}
Z\{(1 - bt)e^{-bt}\} &= (Z(1 - bt))_{z \rightarrow \infty} aT \\
&= (Z(1) - bZ(t))_{z \rightarrow \infty} aT \\
&= \left[\frac{z}{z-1} + bTz \frac{d}{dz} (Z(1)) \right]_{z \rightarrow \infty} aT
\end{aligned}$$

$$\begin{aligned}
&= \left[\frac{z}{z-1} + bTz \frac{d}{dz} \left(\frac{z}{z-1} \right) \right]_{z \rightarrow \infty} aT \\
&= \left[\frac{z}{z-1} + bTz \left(\frac{-1}{(z-1)^2} \right) \right]_{z \rightarrow \infty} aT \\
&= \frac{ze^{bT}}{ze^{bT} - 1} - \frac{bTze^{bT}}{(ze^{bT} - 1)^2} \\
&= \frac{ze^{bT} (ze^{bT} - 1) - bTze^{bT}}{(ze^{bT} - 1)^2} \\
&= \frac{e^{-2bT} ze^{bT} (ze^{bT} - 1 - Tb)}{(z - e^{-bT})^2} \\
&= \frac{z(z - (1 + Tb)e^{-bT})}{(z - e^{-bT})^2}
\end{aligned}$$

(xviii) na^{n-1}

Solution: Since we have,

$$Z(na^n) = \frac{az^2}{z(z-a)^2}$$

Now,

$$Z(na^{n-1}) = a^{-1} Z(na^n) = a^{-1} \left[\frac{az^2}{z(z-a)^2} \right] = \frac{z^2}{z(z-a)^2}$$

(xix) $n^2 a^{n-1}$ for $n \geq 1$

Solution: Since we have,

$$Z(n f(t)) = -z \frac{d}{dz} (Z\{f(t)\}) \quad \text{and} \quad Z\{a^t f(t)\} = (Z\{f(t)\})_{z \rightarrow za}, \quad Z(1) = \frac{z}{z-1}$$

Now, for $n \geq 1$,

$$\begin{aligned}
Z(n^2 a^{n-1}) &= a^{-1} Z(n^2 a^n) \\
&= a^{-1} (Z(n^2))_{z \rightarrow za} \\
&= a^{-1} \left[-z \frac{d}{dz} (Z(n)) \right]_{z \rightarrow za} \\
&= a^{-1} \left[z \frac{d}{dz} \left(z \frac{d}{dz} Z(1) \right) \right]_{z \rightarrow za} \\
&= a^{-1} \left[z \frac{d}{dz} \left(z \frac{d}{dz} \left(\frac{z}{z-1} \right) \right) \right]_{z \rightarrow za} \\
&= a^{-1} \left[z \frac{d}{dz} \left(\frac{-z}{(z-1)^2} \right) \right]_{z \rightarrow za} \\
&= a^{-1} \left[\frac{z[-(z-1)^2 + 2z(z-1)]}{(z-1)^4} \right]_{z \rightarrow za} \\
&= a^{-1} \left[z \left(\frac{-z+1+2z}{(z-1)^3} \right) \right]_{z \rightarrow za} \\
&= a^{-1} \left[\frac{z(z+1)}{(z-1)^3} \right]_{z \rightarrow za}
\end{aligned}$$

Now,

$$\begin{aligned}
 Z(e^{i\pi/3}) &= (Z(1))_{z \rightarrow e^{i\pi/3}} = \left(\frac{z}{z-1}\right)_{z \rightarrow e^{i\pi/3}} \\
 &= \frac{ze^{-i\pi/3}}{ze^{-i\pi/3} - 1} \\
 &= \frac{z}{z - e^{i\pi/3}} \\
 &= \frac{z}{z - \cos \frac{\pi}{3} - i \sin \frac{\pi}{3}} \\
 &= \frac{z}{z - \frac{1}{2} - \frac{i\sqrt{3}}{2}} \\
 &= \frac{2z}{2z - 1 - i\sqrt{3}} \times \frac{2z - 1 + i\sqrt{3}}{2z - 1 + i\sqrt{3}} \\
 &= \frac{2z(2z - 1) + i2\sqrt{3}z}{(2z - 1)^2 + 3} \\
 &= \frac{2z(2z - 1) + i2\sqrt{3}z}{4z^2 - 4z + 4} \\
 &= \frac{z(2z - 1) + iz\sqrt{3}}{2z^2 - 2z + 2}
 \end{aligned}$$

$$\Rightarrow Z\left\{\cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3}\right\} = \frac{z(2z - 1) + iz\sqrt{3}}{2z^2 - 2z + 2}$$

Comparing the real part then,

$$Z\left\{\cos \frac{n\pi}{3}\right\} = \frac{z(2z - 1)}{2z^2 - 2z + 2}$$

(xi) $\sin 4n$ Solution: Since, $\sin 4n = \text{Im}(e^{i4n})$

$$\text{We have, } Z(a^n) = \frac{z}{z - a}$$

Now,

$$\begin{aligned}
 Z(e^{i4n}) &= z[(e^i)^4]^n = \frac{z}{z - e^{i4}} \\
 &= \frac{z}{z - \cos 4 - i \sin 4} \times \frac{z - \cos 4 + i \sin 4}{z - \cos 4 + i \sin 4} \\
 &= \frac{z(z - \cos 4 + i \sin 4)}{(z - \cos 4)^2 + \sin^2 4} \\
 &= \frac{z(z - \cos 4) + iz \sin 4}{z^2 - 2z \cos 4 + 1}
 \end{aligned}$$

$$\Rightarrow Z(\cos 4n + i \sin 4n) = \frac{z(z - \cos 4) + iz \sin 4}{z^2 - 2z \cos 4 + 1}$$

Comparing the imaginary part from both sides then,

$$Z(\sin 4n) = \frac{z \sin 4}{z^2 - 2z \cos 4 + 1}$$

(xii) $\cosh 7n$

Solution: We know that,

$$\cosh 7n = \frac{e^{7n} + e^{-7n}}{2} = \frac{1}{2}[(e^7)^n + (e^{-7})^n]$$

$$\text{Since we have, } Z(a^n) = \frac{z}{z - a}$$

Now,

$$\begin{aligned}
 Z(\cosh 7n) &= \frac{1}{2}[Z((e^7)^n) + Z((e^{-7})^n)] \\
 &= \frac{1}{2}\left[\frac{z}{z - e^7} + \frac{z}{z - e^{-7}}\right] \\
 &= \frac{z}{2}\left[\frac{2z - e^7 - e^{-7}}{(z - e^7)(z - e^{-7})}\right] \\
 &= \frac{z}{2}\left[\frac{2z - 2\cosh 7}{z^2 - z(e^7 + e^{-7}) + e^7 e^{-7}}\right] \\
 &= \frac{z(z - \cosh 7)}{z^2 - 2z \cosh 7 + 1} \quad [\because \cosh 7 = \frac{e^7 + e^{-7}}{2}]
 \end{aligned}$$

Thus,

$$Z(\cosh 7n) = \frac{z(z - \cosh 7)}{z^2 - 2z \cosh 7 + 1}$$

(xiii) $\sinh 8n$

$$\text{Solution: Since, } \sinh 8n = \frac{e^{8n} - e^{-8n}}{2}$$

And the process as in (xii).

(xiv) e^{bn}

$$\text{Solution: Since we have, } Z(a^n) = \frac{z}{z - a}$$

Now,

$$Z(e^{bn}) = Z((e^b)^n) = \frac{z}{z - e^b}$$

(xv) $t^k f(t)$

Solution: Since we have,

$$Z(t^k f(t)) = (-1)^k \left(z \frac{d}{dz}\right)^k [Z(f(t))] \text{ and } Z(a^n) = \frac{z}{z - a}$$

Now,

$$\begin{aligned}
 Z(t^3 e^{-bt}) &= Z(t^3 e^{-bt}) \\
 &= (-1)^3 T\left(z \frac{d}{dz}\right)^3 Z((e^{-b})^n) \\
 &= -T^2\left(z \frac{d}{dz}\right)^2 \left[z \frac{d}{dz} \left(\frac{z}{z - e^{-b}}\right)\right]
 \end{aligned}$$

Solution: Given difference equation is

$$y_{n+2} - y_n = 2^n \quad \dots\dots(i)$$

$$\text{with } y_0 = 0, y_1 = 1 \quad \dots\dots(ii)$$

Now, taking z-transform on (i) then

$$Z\{y_{n+2}\} - Z\{y_n\} = Z\{2^n\}$$

Then,

$$z^2 \left(\bar{y} - y_0 - \frac{y_1}{z} \right) - \bar{y} = \frac{z}{z-2} \quad \left[\because \bar{y} = Z\{y_n\} \text{ and } z\{a^n\} = \frac{z}{z-a} \right]$$

Applying (ii) then

$$z^2 \left(\bar{y} - \frac{1}{z} \right) - \bar{y} = \frac{z}{z-2}$$

$$\Rightarrow \bar{y} (z^2 - 1) - z = \frac{z}{z-2}$$

$$\Rightarrow \bar{y} = \frac{1}{(z^2 - 1)} \left(\frac{z}{z-2} + z \right)$$

$$= \frac{z^2 - z}{(z-2)(z+1)(z-1)}$$

And, taking inverse z-transform then

$$y_n = Z^{-1} \left\{ \frac{z^2 - z}{(z-2)(z+1)(z-1)} \right\} \quad \dots\dots(iii)$$

Let,

$$F(z) = \frac{z^2 - z}{(z-2)(z+1)(z-1)}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^{n+1} - z^n}{(z-2)(z+1)(z-1)}$$

Clearly $z^{n-1} F(z)$ has poles at $z = 2, z = -1$ and $z = 1$ of simple order.

Since we have,

$$z^{-1} \{F(z)\} = \text{sum of residue of } z^{n-1} F(z) \quad \dots\dots(iv)$$

Here, residue of $z^{n-1} F(z)$ at $z = 2$ is

$$\begin{aligned} \text{Res}_{z=2} (z^{n-1} F(z)) &= \lim_{z \rightarrow 2} [(z-2) z^{n-1} F(z)] \\ &= \lim_{z \rightarrow 2} \left(\frac{z^{n+1} - z^n}{(z+1)(z-1)} \right) = \frac{2^{n+1} - 2^n}{(3)(1)} = \frac{2^n(2-1)}{3} = \frac{2^n}{3} \end{aligned}$$

And residue of $z^{n-1} F(z)$ at $z = -1$ is,

$$\begin{aligned} \text{Res}_{z=-1} (z^{n-1} F(z)) &= \lim_{z \rightarrow -1} [(z+1) z^{n-1} F(z)] \\ &= \lim_{z \rightarrow -1} \left(\frac{z^{n+1} - z^n}{(z-2)(z-1)} \right) \\ &= \frac{(-1)^{n+1} - (-1)^n}{(-3)(-2)} = \frac{(-1)^n(-1-1)}{(-3)(-2)} = \frac{(-1)^{n+1}}{3} \end{aligned}$$

Also, residue of $z^{n-1} F(z)$ at $z = 1$ is,

$$\begin{aligned} \text{Res}_{z=1} (z^{n-1} F(z)) &= \lim_{z \rightarrow 1} [(z-1) z^{n-1} F(z)] \\ &= \lim_{z \rightarrow 1} \left(\frac{z^{n+1} - z^n}{(z-2)(z+1)} \right) = \frac{1^{n+1} - 1^n}{(-1)(2)} = 0 \end{aligned}$$

Thus, with (iv), the equation (iii) becomes,

$$\begin{aligned} y_n &= \frac{2^n}{3} + \frac{(-1)^{n+1}}{3} + 0 \\ \Rightarrow y_n &= \frac{2^n + (-1)^{n+1}}{3} \end{aligned}$$

This is required solution of (i) satisfying (ii).

$$(v) \quad y_{x+2} - 3y_{x+1} + 2y_x = 0 \quad \text{with } y_0 = 0, y_1 = 1.$$

[2006 Fall Q. No. 5(a)]

Solution: Given difference equation is,

$$y_{x+2} - 3y_{x+1} + 2y_x = 0 \quad \dots\dots(i)$$

$$\text{with } y_0 = 0, y_1 = 1 \quad \dots\dots(ii)$$

Now, taking z-transform on (i) then,

$$Z\{y_{x+2}\} - 3Z\{y_{x+1}\} + 2Z\{y_x\} = Z\{0\}$$

Then,

$$z^2 \left(\bar{y} - y_0 - \frac{y_1}{z} \right) - 3z(\bar{y} - y_0) + 2\bar{y} = 0 \quad \text{where, } \bar{y} = Z\{y_x\}$$

Applying (ii) then,

$$\bar{y} (z^2 - 3z + 2) - z = 0$$

$$\Rightarrow \bar{y} = \frac{z}{z^2 - 3z + 2}$$

And taking inverse z-transform we get,

$$y_n = Z^{-1} \left\{ \frac{z}{z^2 - 3z + 2} \right\} \quad \dots\dots(iii)$$

$$\text{Let, } F(z) = \frac{z}{z^2 - 3z + 2}$$

Then, [by Q. No. 2(vi)]

Thus, with (iv), the equation (iii) becomes,

$$y_n = 1 - 2^n.$$

$$(vi) \quad y_{x+2} - 4y_{x+1} + 4y_x = 0 \quad \text{with } y_0 = 1, y_1 = 0.$$

[2009 Fall Q. No. 6(b) OR] [2008 Fall Q. No. 6(b) OR] [2004 Spring Q. No. 3(b)]

Solution: Given different equation is,

$$y_{x+2} - 4y_{x+1} + 4y_x = 0 \quad \dots\dots(1)$$

$$\text{with } y_0 = 1, y_1 = 0 \quad \dots\dots(2)$$

Now, taking z-transform on (i) then,

$$Z\{y_{x+2}\} - 4Z\{y_{x+1}\} + 4Z\{y_x\} = Z\{0\}$$

Then,

$$\begin{aligned}
 &= \lim_{z \rightarrow 1} \frac{d}{dz} \left[\frac{z^n}{(z+1)^2} \right] \\
 &= \lim_{z \rightarrow 1} \left[\frac{nz^{n-1}(z+1)^2 - 2z^n(z+1)}{(z+1)^4} \right] \\
 &= \frac{n(1)^{n-1}(2)^2 - 2(1)^n(2)}{(2)^4} \\
 &= \frac{4n-4}{16} = \frac{n-1}{4}
 \end{aligned}$$

And, residue of $z^{n-1} F(z)$ at $z = -1$ of order 2 is,

$$\begin{aligned}
 \text{Res}_{z=-1} z^{n-1} F(z) &= \lim_{z \rightarrow -1} \frac{d}{dz} [(z+1)^2 z^{n-1} F(z)] \\
 &= \lim_{z \rightarrow -1} \frac{d}{dz} \left[\frac{z^n}{(z-1)^2} \right] \\
 &= \lim_{z \rightarrow -1} \left[\frac{nz^{n-1}(z-1)^2 - 2z^n(z-1)}{(z-1)^4} \right] \\
 &= \frac{n(-1)^{n-1}(-2)^2 - 2(-1)^n(-2)}{(-2)^4} \\
 &= \frac{4n(-1)^{n-1} + 4(-1)^n}{16} \\
 &= \frac{(-1)^{n-1}(n-1)}{4}
 \end{aligned}$$

Thus, from (iv) and (v) we get,

$$y_n = \left(\frac{n-1}{4} \right) (1 + (-1)^{n-1})$$

$$\Rightarrow y_n = \left(\frac{n-1}{4} \right) [1 - (-1)^n]$$

This is the required solution of (i) satisfying (ii).

(iii) $u_{x+2} - 2 \cos \alpha u_{x+1} + u_x = 0$ with $u_0 = 1, u_1 = \cos \alpha$

Solution: Given difference equation is

$$u_{x+2} - 2 \cos \alpha u_{x+1} + u_x = 0 \quad \dots\dots(i)$$

with $u_0 = 1, u_1 = \cos \alpha$

Now, taking z-transform on (i) then

$$Z\{u_{x+2}\} - 2 \cos \alpha Z\{u_{x+1}\} + Z\{u_x\} = Z\{0\}$$

Then,

$$z^2 \left(\bar{U} - U_0 - \frac{U_1}{z} \right) - 2z \cos \alpha (\bar{U} - U_0) + \bar{U} = 0 \quad [\because Z\{u_x\} = \bar{U}]$$

Applying (ii) then,

$$\bar{U} [z^2 - 2z \cos \alpha + 1] - z^2 - z \cos \alpha + 2z \cos \alpha = 0$$

$$\Rightarrow \bar{U} = \frac{z^2 - z \cos \alpha}{z^2 - 2z \cos \alpha + 1}$$

Now taking inverse z-transform then,

$$\begin{aligned}
 U_x &= Z^{-1} \left\{ \frac{z^2 - z \cos \alpha}{z^2 - 2z \cos \alpha + 1} \right\} \quad \dots\dots(iii) \\
 &= \frac{z^2 - z \cos \alpha}{(z - e^{i\alpha})(z - e^{-i\alpha})}
 \end{aligned}$$

Then,

$$z^{x-1} F(z) = \frac{z^{x+1} - z^x \cos \alpha}{(z - e^{i\alpha})(z - e^{-i\alpha})}$$

Clearly, $z^{x-1} F(z)$ has poles at $z = e^{i\alpha}$ of order 1 and at $z = e^{-i\alpha}$ of order 1.

Since we have,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{x-1} F(z) \quad \dots\dots(iv)$$

Here, residue of $z^{x-1} F(z)$ at $z = e^{i\alpha}$ is

$$\begin{aligned}
 \text{Res}_{z=e^{i\alpha}} z^{x-1} F(z) &= \lim_{z \rightarrow e^{i\alpha}} [(z - e^{i\alpha}) z^{x-1} F(z)] \\
 &= \lim_{z \rightarrow e^{i\alpha}} \frac{z^{x+1} - z^x \cos \alpha}{z - e^{-i\alpha}} \\
 &= \frac{e^{i\alpha(x+1)} - e^{i\alpha x} \cos \alpha}{e^{i\alpha} - e^{-i\alpha}}
 \end{aligned}$$

And the residue of $z^{x-1} F(z)$ at $z = e^{-i\alpha}$ is obtained from above with replacing i by $-i$ as,

$$\text{Res}_{z=e^{-i\alpha}} [z^{x-1} F(z)] = \frac{e^{-i\alpha(x+1)} - e^{-i\alpha x} \cos \alpha}{-2i \sin \alpha}$$

Thus, (iii) and (iv) gives,

$$\begin{aligned}
 U_x &= \frac{e^{i\alpha(x+1)} - e^{i\alpha x} \cos \alpha}{2i \sin \alpha} + \frac{e^{-i\alpha(x+1)} - e^{-i\alpha x} \cos \alpha}{-2i \sin \alpha} \\
 &= \frac{e^{i\alpha(x+1)} - e^{-i\alpha(x+1)} - e^{i\alpha x} \cos \alpha + e^{-i\alpha x} \cos \alpha}{2i \sin \alpha} \\
 &= \frac{2i \sin (x+1)\alpha - 2i \sin \alpha x \cos \alpha}{2i \sin \alpha} \\
 &= \frac{\sin \alpha x \cos \alpha + \cos \alpha x \sin \alpha - \sin \alpha x \cos \alpha}{\sin \alpha} = \frac{\cos \alpha x \sin \alpha}{\sin \alpha} = \cos \alpha x
 \end{aligned}$$

This is the required solution of (i) satisfying (ii).

(iv) $y_{n+2} - y_n = 2^n$ with $y_0 = 0, y_1 = 1$.

[2005 Spring Q. No. 3(b)] [2009 Spring Q. No. 3(b)]

$$F(z) = \frac{20z + 5}{(z-2)(z-1/2)}$$

$$\text{Then, } z^{n-1} F(z) = \frac{20z^n + 5z^{n-1}}{(z-2)(z-1/2)}$$

Clearly $z^{n-1} F(z)$ has simple poles at $z = 2$ and $z = 1/2$.

Here, residue of $z^{n-1} F(z)$ at $z = 2$ is,

$$\begin{aligned} \text{Res}_{z=2} z^{n-1} F(z) &= \lim_{z \rightarrow 2} \left(\frac{20z^n + 5z^{n-1}}{z-1/2} \right) \\ &= \frac{20(2)^n + 5(2)^{n-1}}{3/2} = 2^n \left(20 + \frac{5}{2} \right) \times \frac{2}{3} = 15(2)^n \end{aligned}$$

And, residue of $z^{n-1} F(z)$ at $z = 1/2$ is,

$$\begin{aligned} \text{Res}_{z=1/2} z^{n-1} F(z) &= \lim_{z \rightarrow 1/2} \left(\frac{20z^n + 5z^{n-1}}{z-2} \right) \\ &= \frac{20(1/2)^n + 5(1/2)^{n-1}}{-3/2} \\ &= \left(\frac{1}{2} \right)^n (20 + 10) \times \left(-\frac{2}{3} \right) = -20 \left(\frac{1}{2} \right)^n \end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = 15(2)^n - 20 \left(\frac{1}{2} \right)^n$$

3. Solve by using Z-transform:

$$(i) \quad y_{n+2} + y_n = 1 \text{ with } y_0 = y_1 = 0.$$

Solution: Given difference equation is

$$y_{n+2} + y_n = 1 \quad \dots\dots(i)$$

$$\text{with } y_0 = y_1 = 0 \quad \dots\dots(ii)$$

Since we have,

$$Z\{y_{n+k}\} = z^k \left[\bar{y} - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2} - \dots - \frac{y_{k-1}}{z^{k-1}} \right] \quad \text{where, } \bar{y} = Z\{y_n\}$$

Now, taking z-transform on (i) then,

$$Z\{y_{n+2}\} + Z\{y_n\} = Z\{1\}$$

$$\Rightarrow z^2 \left[\bar{y} - y_0 - \frac{y_1}{z} \right] + \bar{y} = \frac{z}{z-1} \quad \left[\because \bar{y} = Z\{y_n\} \text{ and } z\{1\} = \frac{z}{z-1} \right]$$

Applying (ii) then,

$$z^2 \bar{y} + \bar{y} = \frac{z}{z-1}$$

$$\Rightarrow \bar{y}(z^2 + 1) = \frac{z}{z-1}$$

$$\Rightarrow \bar{y} = \frac{z}{(z-1)(z^2+1)}$$

Now, taking inverse z-transform we get,

$$y_n = Z^{-1} \left\{ \frac{z}{(z-1)(z^2+1)} \right\} \quad \dots\dots(iii)$$

Therefore (iii) becomes,

$$y_n = \frac{1}{2} \left[1 - \left(\cos \left(\frac{n\pi}{2} \right) + \sin \left(\frac{n\pi}{2} \right) \right) \right]$$

This is required solution of (i) satisfying (ii).

$$(ii) \quad y_{n+2} + 2y_{n+1} + y_n = n \text{ with } y_0 = y_1 = 0.$$

[2003 Fall Q. No. 5(a)] [2006 Spring Q. No. 5(a)]

Solution: Given difference equation is,

$$y_{n+2} + 2y_{n+1} + y_n = n \quad \dots\dots(i)$$

$$\text{with } y_0 = y_1 = 0 \quad \dots\dots(ii)$$

Since we have

$$Z\{y_{n+k}\} = z^k \left[\bar{y} - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2} - \dots - \frac{y_{k-1}}{z^{k-1}} \right] \quad \text{where, } \bar{y} = Z\{y_n\}$$

Now, taking z-transform on (i) then

$$Z\{y_{n+2}\} + 2Z\{y_{n+1}\} + Z\{y_n\} = Z\{n\}$$

Then,

$$z^2 \left(\bar{y} - y_0 - \frac{y_1}{z} \right) + 2z(\bar{y} - y_0) + \bar{y} = \frac{z}{(z-1)^2} \quad \dots\dots(iii)$$

$$\left[\because \bar{y} = Z\{y_n\} \text{ and } Z\{n\} = \frac{z}{(z-1)^2} \right]$$

Applying (ii) on (iii) then

$$\bar{y}(z^2 + 2z + 1) = \frac{z}{(z-1)^2}$$

$$\Rightarrow \bar{y} = \frac{z}{(z-1)^2(z^2+2z+1)} = \frac{z}{(z-1)^2(z+1)^2}$$

Now, taking inverse z-transform then

$$y_n = Z^{-1} \left\{ \frac{z}{(z-1)^2(z+1)^2} \right\} \quad \dots\dots(iv)$$

$$\text{Let, } F(z) = \frac{z}{(z-1)^2(z+1)^2}$$

Then,

$$z^{n-1} F(z) = \frac{z^n}{(z-1)^2(z+1)^2}$$

Clearly $z^{n-1} F(z)$ has poles at $z = 1$ of order 2 and at $z = -1$ of order 2.

Since we have,

$$z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z) \quad \dots\dots(v)$$

Here, residue of $z^{n-1} F(z)$ at $z = 1$ of order 2 is,

$$\text{Res}_{z=1} z^{n-1} F(z) = \lim_{z \rightarrow 1} \frac{d}{dz} [(z-1)^2 z^{n-1} F(z)]$$

Here, residue of $z^{n-1} F(z)$ at $z = -1$ is,

$$\begin{aligned}\text{Res}_{z=-1} z^{n-1} F(z) &= \lim_{z \rightarrow -1} (z+1) \left(\frac{z^{n+1} + 6z^{n-1} + z^{n-2}}{(z+1) \left(z + \frac{1}{5} \right)} \right) \\ &= \frac{(-1)^{n+1} + 6(-1)^{n-1} + (-1)^{n-2}}{-4/5} \\ &= (-1)^n (-1 - 6 + 1) \times \frac{5}{4} = (-1)^n \frac{3 \times 5}{2} = \frac{15}{2} (-1)^n\end{aligned}$$

And, residue of $z^{n-1} F(z)$ at $z = -\frac{1}{5}$ is,

$$\begin{aligned}\text{Res}_{z=-\frac{1}{5}} z^{n-1} F(z) &= \lim_{z \rightarrow -\frac{1}{5}} \left(\frac{z^{n+1} + 6z^{n-1} + z^{n-2}}{(z+1)} \right) \\ &= \frac{\left(-\frac{1}{5}\right)^{n+1} + 6\left(-\frac{1}{5}\right)^{n-1} + \left(-\frac{1}{5}\right)^{n-2}}{\left(-\frac{1}{5} + 1\right)} \\ &= \frac{\left(-\frac{1}{5}\right)^n \left(\left(-\frac{1}{5}\right)^1 + 6\left(-\frac{1}{5}\right)^{-1} + \left(-\frac{1}{5}\right)^{-2} \right)}{\frac{4}{5}} \\ &= (-1)^n \left(\frac{1}{5} \right)^n \left(-\frac{1}{5} - 30 + 25 \right) \times \frac{5}{4} \\ &= -(-1)^n \left(\frac{1}{5} \right)^n \frac{13}{2}\end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = f(t) = \frac{15}{2} (-1)^n - \frac{13}{2} (-1)^n \left(\frac{1}{5} \right)^n$$

$$(xxx) \frac{z-4}{(z-1)(z-2)^2}$$

$$\text{Solution: Let, } F(z) = \frac{z-4}{(z-1)(z-2)^2}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^n - 4z^{n-1}}{(z-1)(z-2)^2}$$

Clearly $z^{n-1} F(z)$ has simple poles at $z = 1$ and $z = 2$ of order 2.

Here, residue of $z^{n-1} F(z)$ at $z = 1$ is,

$$\text{Res}_{z=1} z^{n-1} F(z) = \lim_{z \rightarrow 1} \frac{z^n - 4z^{n-1}}{(z-2)^2} = 1 - 4 = -3$$

And, residue of $z^{n-1} F(z)$ at $z = 2$ is,

$$\begin{aligned}\text{Res}_{z=2} z^{n-1} F(z) &= \lim_{z \rightarrow 2} \frac{d}{dz} \left[\frac{z^n - 4z^{n-1}}{z-1} \right] \\ &= \lim_{z \rightarrow 2} \left\{ \frac{n z^{n-1} - 4(n-1) z^{n-2}}{(z-1)^2} (z-1) - \frac{z^n - 4z^{n-1}}{(z-1)^2} \right\} \\ &= n \cdot 2^{n-1} - 4n \cdot 2^{n-2} + 4 \cdot 2^{n-2} - 2^n + 4 \cdot (2)^{n-1} \\ &= 2^{n-1} \left\{ n - 4n \cdot \frac{1}{2} + 4 \cdot \frac{1}{2} - 2 + 4 \right\} \\ &= 2^{n-1} (n - 2n + 4) \\ &= -2^{n-1} (n - 4)\end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = -3 - 2^{n-1} (n - 4)$$

$$(xxxi) \frac{2z}{(2z-1)^2}$$

Solution: Let,

$$F(z) = \frac{2z}{(2z-1)^2} = \frac{z}{2(z-1/2)^2}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^n}{2(z-1/2)^2}$$

Clearly $z^{n-1} F(z)$ has pole at $z = 1/2$ of order 2.

Here, residue of $z^{n-1} F(z)$ at $z = \frac{1}{2}$ is,

$$\begin{aligned}\text{Res}_{z=1/2} z^{n-1} F(z) &= \lim_{z \rightarrow 1/2} \frac{d}{dz} \left[\frac{z^n}{2(z-1/2)^2} \times (z-1/2)^2 \right] \\ &= \frac{n z^{n-1}}{2} \\ &= \frac{n (1/2)^{n-1}}{2} = n \left(\frac{1}{2} \right)^n\end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = n \left(\frac{1}{2} \right)^n$$

$$(xxxi) \frac{z^3 - 20z}{(z-4)(z-2)^3}$$

Solution: Process as in (xxiv).

$$(xxxiii) \frac{20z+5}{(z-2)(z-1/2)}$$

Solution: Let,

$$= \frac{\left(\frac{1}{2} + \frac{i}{2}\right)^{n+1} + \left[\left(\frac{1}{2} + \frac{i}{2}\right)^n + \left(\frac{1}{2} + \frac{i}{2}\right)\right]}{i}$$

$$= -\left(\frac{1}{2}\right)^{n+1} i(1+i)^{n+1} - \left(\frac{1}{2}\right)^n \cdot i(1+i)^n$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

Therefore,

$$Z^{-1}\{F(z)\} = \left(\frac{1}{2}\right)^{n+1} i[(1-i)^{n+1} - (1+i)^{n+1}] + \left(\frac{1}{2}\right)^n i[(1-i)^n - (1+i)^n]$$

$$(xxvii) \frac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

Solution: Here,

$$F(z) = \frac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} = \frac{z^2}{z^2 - \frac{3}{2}z + \frac{1}{2}}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^{n+1}}{z^2 - \frac{3}{2}z + \frac{1}{2}} = \frac{z^{n+1}}{(z-1)\left(z - \frac{1}{2}\right)}$$

Clearly $z^{n-1} F(z)$ has simple poles at $z = 1$ and $z = \frac{1}{2}$.

Here, residue of $z^{n-1} F(z)$ at $z = 1$ of order 1 is,

$$\text{Res}_{z=1} z^{n-1} F(z) = \lim_{z \rightarrow 1} (z-1) \frac{z^{n+1}}{(z-1)\left(z - \frac{1}{2}\right)} = \frac{1}{1/2} = 2$$

And, residue of $z^{n-1} F(z)$ at $z = \frac{1}{2}$ of order 1 is,

$$\text{Res}_{z=1/2} z^{n-1} F(z) = \lim_{z \rightarrow 1/2} \left(\frac{z^{n+1}}{z-1}\right) = \frac{(1/2)^{n+1}}{-1/2} = -\left(\frac{1}{2}\right)^n$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = 2 - \left(\frac{1}{2}\right)^n$$

$$(xxviii) \frac{16z^3}{(4z-1)^2(z-1)}$$

Solution: Let,

$$F(z) = \frac{16z^3}{(4z-1)^2(z-1)} = \frac{z^3}{(z-1/4)^2(z-1)}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^{n+2}}{(z-1/4)^2(z-1)}$$

Clearly $z^{n-1} F(z)$ has simple poles at $z = 1$ and at $z = \frac{1}{4}$ of order 2.

Here, residue of $z^{n-1} F(z)$ at $z = 1$ is,

$$\text{Res}_{z=1} z^{n-1} F(z) = \lim_{z \rightarrow 1} \frac{z^{n+2}}{(z-1/4)^2} = \frac{1}{(3/4)^2} = \frac{16}{9}$$

Here, residue of $z^{n-1} F(z)$ at $z = 1/4$ is,

$$\begin{aligned} \text{Res}_{z=1/4} z^{n-1} F(z) &= \lim_{z \rightarrow 1/4} \frac{d}{dz} \left(\frac{z^{n+2}}{z-1} \right) \\ &= \lim_{z \rightarrow 1/4} \left\{ \frac{(n+2)z^{n+1}(z-1) - z^{n+2}}{(z-1)^2} \right\} \\ &= 2 \left(\frac{1}{4}\right)^3 - 3 \left(\frac{1}{4}\right)^2 \\ &= \frac{(n+2) \left(\frac{1}{4}\right)^{n+1} \times \frac{-3}{4} - \left(\frac{1}{4}\right)^{n+2}}{\frac{9}{16}} \\ &= \left(\frac{1}{4}\right)^{n+2} \{-3n-6-1\} \times \frac{16}{9} \\ &= -\left(\frac{1}{4}\right)^n \frac{1}{9} (3n+7) \end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = f(t) = \frac{16}{9} - \left(\frac{1}{4}\right)^n \frac{1}{9} (3n+7).$$

$$(xix) \frac{1+6z^{-2}+z^{-3}}{(1+z^{-1})\left(1+\frac{1}{5}z^{-1}\right)}$$

Solution: Let,

$$\begin{aligned} F(z) &= \frac{1+6z^{-2}+z^{-3}}{(1+z^{-1})\left(1+\frac{1}{5}z^{-1}\right)} \\ &= \frac{z^{-3}(z^3+6z+1)}{z^{-2}(z+1)\left(z+\frac{1}{5}\right)} = \frac{z^{-1}(z^3+6z+1)}{(z+1)\left(z+\frac{1}{5}\right)} \end{aligned}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^{n+1}+6z^{n-1}+z^{n-2}}{(z+1)\left(z+\frac{1}{5}\right)}$$

Clearly $z^{n-1} F(z)$ has simple poles at $z = -1$ and $z = -\frac{1}{5}$.

$$\begin{aligned}
 &= \lim_{z \rightarrow 1/3} \left\{ \frac{(n+2)z^{n+1}(z-2) - z^{n+2}}{(z-2)^2} \right\} \\
 &= \frac{(n+2) \left(\frac{1}{3}\right)^{n+1} \left(\frac{1}{3} - 2\right) - \left(\frac{1}{3}\right)^{n+2}}{\left(\frac{1}{3} - 2\right)^2} \\
 &= \frac{(n+2) \left(\frac{1}{3}\right)^{n+1} \left(-\frac{5}{3}\right) - \left(\frac{1}{3}\right)^{n+2}}{\frac{25}{9}} \\
 &= \left(\frac{1}{3}\right)^{n+2} [(n+2) \times -5 - 1] \times \frac{9}{25} \\
 &= -\left(\frac{1}{3}\right)^n \left\{ \frac{(n+2)}{5} + \frac{1}{25} \right\}
 \end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1}F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = \frac{9}{25} 2^{n+2} - \left(\frac{1}{3}\right)^n \left\{ \frac{(n+2)}{5} + \frac{1}{25} \right\}$$

$$(xxv) \frac{2z^2 + 3z}{z^2 + z + 1}$$

Solution: Let,

$$F(z) = \frac{2z^2 + 3z}{z^2 + z + 1}$$

$$\begin{aligned}
 \text{Then, } z^{n-1}F(z) &= \frac{2z^{n+1} + 3z^n}{\left(z + \frac{1}{2} - i\frac{\sqrt{3}}{2}\right)\left(z + \frac{1}{2} + i\frac{\sqrt{3}}{2}\right)} \\
 &= \frac{2z^{n+1} + 3z^n}{(z - e^{i(2\pi/3)})(z - e^{-i(2\pi/3)})}
 \end{aligned}$$

Clearly $z^{n-1}F(z)$ has poles Here, residue of $z^{n-1}F(z)$ at $z = e^{i(2\pi/3)}$ and $e^{-i(2\pi/3)}$ which are simple.

Here, residue of $z^{n-1}F(z)$ at $z = e^{i(2\pi/3)}$ is,

$$\begin{aligned}
 \text{Res}_{z=e^{i(2\pi/3)}} z^{n-1}F(z) &= \lim_{z \rightarrow e^{i(2\pi/3)}} \frac{2z^{n+1} + 3z^n}{z - e^{-i(2\pi/3)}} \\
 &= \frac{2e^{i(n+1)(2\pi/3)} + 3e^{i(2n\pi/3)}}{e^{i(2\pi/3)} - e^{-i(2\pi/3)}} \\
 &= \frac{e^{i2n\pi/3}(2e^{i2\pi/3} + 3)}{\sqrt{3}i}
 \end{aligned}$$

And, similarly the residue of $z^{n-1}F(z)$ at $z = e^{-i(2\pi/3)}$ is,

$$\text{Res}_{z=e^{-i(2\pi/3)}} z^{n-1}F(z) = \frac{e^{-i2n\pi/3}(2e^{-i2\pi/3} + 3)}{-\sqrt{3}i}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1}F(z)$$

$$\begin{aligned}
 \text{Therefore, } Z^{-1}\{F(z)\} &= \frac{e^{i2n\pi/3}(2e^{i2\pi/3} + 3)}{\sqrt{3}i} + \frac{e^{-i2n\pi/3}(2e^{-i2\pi/3} + 3)}{-\sqrt{3}i} \\
 &= e^{i2n\pi/3} \frac{(2 + \sqrt{3}i)}{\sqrt{3}i} + e^{-i2n\pi/3} \frac{(2 - \sqrt{3}i)}{-\sqrt{3}i} \\
 &= \frac{2}{\sqrt{3}i} (e^{i2n\pi/3} - e^{-i2n\pi/3}) + (e^{i2n\pi/3} + e^{-i2n\pi/3}) \\
 &= \frac{4}{\sqrt{3}} \sin \frac{2n\pi}{3} + 2 \cos \frac{2n\pi}{3}
 \end{aligned}$$

$$(xxvi) \frac{1+z^{-1}}{1-z^{-1} + \frac{1}{2}z^{-2}}$$

$$\text{Solution: Let, } F(z) = \frac{1+z^{-1}}{1-z^{-1} + \frac{1}{2}z^{-2}} = \frac{z(z+1)}{z^2 - z + \frac{1}{2}}$$

$$\Rightarrow F(z) = \frac{z^2 + z}{\left(z - \frac{1}{2}\right)^2 - \frac{1}{4}} = \frac{z^2 + z}{\left(z - \frac{1}{2}\right)^2 - \frac{1}{4}}$$

$$\Rightarrow z^{n-1}F(z) = \frac{z^{n+1} + z^n}{\left(z - \frac{1}{2} + \frac{i}{2}\right)\left(z - \frac{1}{2} - \frac{i}{2}\right)}$$

Clearly $z^{n-1}F(z)$ has simple poles $\frac{1}{2} - \frac{i}{2}$ and $\frac{1}{2} + \frac{i}{2}$

Here, residue of $z^{n-1}F(z)$ at $z = \frac{1}{2} - \frac{i}{2}$ of order 1 is,

$$\begin{aligned}
 \text{Res}_{z=\frac{1}{2}-\frac{i}{2}} z^{n-1}F(z) &= \lim_{z \rightarrow \frac{1}{2}-\frac{i}{2}} \frac{z^{n-1} \cdot (z^2 + z)}{z - \frac{1}{2} - \frac{i}{2}} \\
 &= \frac{\left(\frac{1}{2} - \frac{i}{2}\right)^{n+1} + \left(\frac{1}{2} - \frac{i}{2}\right)^n}{-i} \\
 &= \left(\frac{1}{2}\right)^{n+1} i(1-i)^{n+1} + \left(\frac{1}{2}\right)^n \cdot i(1-i)^n
 \end{aligned}$$

And, residue of $z^{n-1}F(z)$ at $z = \frac{1}{2} + \frac{i}{2}$ of order 1 is,

$$\text{Res}_{z=\frac{1}{2}+\frac{i}{2}} z^{n-1}F(z) = \lim_{z \rightarrow \frac{1}{2}+\frac{i}{2}} \frac{z^{n-1} \cdot (z^2 + z)}{z - \frac{1}{2} + \frac{i}{2}}$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = 4 + \frac{\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^{n-1} \left[\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^2 + \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) + 2\right]}{\frac{\sqrt{3}}{2}i - \left(\frac{\sqrt{3}}{2}i - \frac{1}{2}\right)} + \frac{\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)^{n-1} \left[\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)^2 + \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) + 2\right]}{-\frac{\sqrt{3}}{2}i - \left(\frac{\sqrt{3}}{2}i - \frac{1}{2}\right)}$$

$$\text{Since, } \frac{1}{2} + \frac{\sqrt{3}}{2}i = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$$

$$\text{and } \frac{1}{2} - \frac{\sqrt{3}}{2}i = \cos \frac{\pi}{3} - i \sin \frac{\pi}{3}$$

and process for book answer.

$$(xxii) \frac{z^3}{(1-z^{-1})\left(1-\frac{1}{5}z^{-1}\right)}$$

Solution: Let,

$$F(z) = \frac{z^3}{(1-z^{-1})\left(1-\frac{1}{5}z^{-1}\right)} = \frac{z^3}{z^{-2}(z-1)(z-1/5)} = \frac{z^{-1}}{(z-1)(z-1/5)}$$

Then,

$$z^{n-1}F(z) = \frac{z^{n-2}}{(z-1)(z-1/5)}$$

Clearly $z^{n-1}F(z)$ has poles Here poles are $z = 1, \frac{1}{5}$

Here, residue of $z^{n-1}F(z)$ at $z = 1$ is,

$$\text{Res}_{z=1} z^{n-1}F(z) = \lim_{z \rightarrow 1} \frac{z^{n-2}}{(z-1/5)} = \frac{1}{4/5} = \frac{5}{4}$$

And, residue of $z^{n-1}F(z)$ at $z = 1/5$ is,

$$\begin{aligned} \text{Res}_{z=1/5} z^{n-1}F(z) &= \lim_{z \rightarrow 1/5} \left(\frac{z^{n-2}}{z-1} \right) \\ &= \frac{(1/5)^{n-2}}{-4/5} = \left(\frac{1}{5}\right)^n \times -\frac{5}{4} \times 25 = -\frac{125}{4} \left(\frac{1}{5}\right)^n \end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1}F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = \frac{5}{4} - \frac{125}{4} \left(\frac{1}{5}\right)^n$$

$$(xxiii) \frac{z(3z^2-6z+4)}{(z-1)^2(z-2)}$$

Solution: Let,

$$F(z) = \frac{z(3z^2-6z+4)}{(z-1)^2(z-2)}$$

Then,

$$z^{n-1}F(z) = \frac{3z^{n+2}-6z^{n+1}+4z^n}{(z-1)^2(z-2)}$$

Clearly $z^{n-1}F(z)$ has poles at $z = 1$ of degree 2 and $z = 2$ of degree 1.

Here, residue of $z^{n-1}F(z)$ at $z = 2$ is,

$$\begin{aligned} \text{Res}_{z=2} z^{n-1}F(z) &= \lim_{z \rightarrow 2} \frac{3z^{n+2}-6z^{n+1}+4z^n}{(z-1)^2} \\ &= 3(2)^{n+2} - 6(2)^{n+1} + 4(2)^n \\ &= 2^n(12 - 12 + 4) \\ &= 4 \cdot 2^n \\ &= 2^{n+2} \end{aligned}$$

And,

$$\begin{aligned} \text{Res}_{z=1} z^{n-1}F(z) &= \lim_{z \rightarrow 1} \frac{d}{dz} \left((z-1)^2 \frac{3z^{n+2}-6z^{n+1}+4z^n}{(z-1)^2(z-2)} \right) \\ &= \lim_{z \rightarrow 1} \frac{\{3(n+2)z^{n+1} - 6(n+1) + n4z^{n-1}\}(z-2) - 3z^{n+2} + 6z^{n+1} - 4z^n}{(z-2)^2} \\ &= \{3(n+2) - 6(n+1) + 4n\} \times (-1) - 3 + 6 - 4 \\ &= -3n - 6 + 6n + 6 - 4n - 1 \\ &= -n - 1 \end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1}F(z)$$

$$\text{Therefore, } Z^{-1}\{F(z)\} = 2^{n+2} - n - 1$$

$$(xxiv) \frac{9z^3}{(3z-1)^2(z-2)}$$

Solution: Let,

$$F(z) = \frac{9z^3}{(3z-1)^2(z-2)} = \frac{z^3}{((z-1/3)^2)(z-2)}$$

$$\text{Then, } z^{n-1}F(z) = \frac{z^{n+2}}{(z-1/3)^2(z-2)}$$

Clearly, $z^{n-1}F(z)$ has simple poles at $z = 2$ and $z = \frac{1}{3}$ of degree 2.

Here, residue of $z^{n-1}F(z)$ at $z = 2$ is,

$$\text{Res}_{z=2} z^{n-1}F(z) = \lim_{z \rightarrow 2} \frac{z^{n+2}}{(z-1/3)^2} = \frac{2^{n+2}}{(5/3)^2} = \frac{9}{2^2} (2^{n+2})$$

And, residue of $z^{n-1}F(z)$ at $z = \frac{1}{3}$ is,

$$\text{Res}_{z=1/3} z^{n-1}F(z) = \text{Res}_{z=1/3} \frac{d}{dz} \left(\left(z - \frac{1}{3} \right)^2 \frac{z^{n+2}}{(z-2)(z-1/3)} \right)$$

$$\begin{aligned}
 &= \frac{2i}{(2i)^4} [2ni^{n+2} + 4i^{n+2} - 2i^{n+2} + 2i^2] \\
 &= \frac{2i}{8} [ni^{n+2} + i^{n+2} + i^2] \\
 &= \frac{i^{n+1}}{4} (ni^2 + i^2 + 1) = \frac{i^{n+1}}{4} (-n-1+1) = -\frac{n(i^{n+1})}{4}
 \end{aligned}$$

And residue of $z^{n-1} F(z)$ at $z = -i$ of order 2 be,

$$\begin{aligned}
 \text{Res}_{z=-i} z^{n-1} F(z) &= \lim_{z \rightarrow -i} \frac{d}{dz} \{z^{n-1} F(z) (z+i)^2\} \\
 &= \lim_{z \rightarrow -i} \frac{d}{dz} \left\{ \frac{z^{n+2} - z^n}{(z-i)^4} \right\} \\
 &= \lim_{z \rightarrow -i} \frac{(z-i)^2 \{(n+2)z^{n+1} - nz^{n-1}\} + (z^{n+2} - z^n) 2(z-i)}{(z-i)^4} \\
 &= \frac{(-2i)^2 \{(n+2)(-i)^{n+1} - n(-i)^{n-1}\} - [(-i)^{n+2} - (-i)^n] 2(-2i)}{(-2i)^4} \\
 &= \frac{(-2i)^2 (-i)^{n-1} [2(n+2)(-i)^3 - n(-i) \cdot 2 - 2(-i)^3 + 2(-i)]}{16} \\
 &= \frac{(-i)^n [2n(-i)^3 + 4(-i)^3 - 2n(-i) - 2(-i)^3 + 2(-i)]}{8} \\
 &= \frac{(-i)^n}{8} (2n \cdot i + 2(-i)^3 + 2ni - 2i) \\
 &= \frac{(-i)^n}{8} (4ni - 2i - 2i) \\
 &= \frac{(-i)^n}{8} (4ni - 4i) = \frac{(-i)^{n+1}}{2} (1-n)
 \end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$

$$\begin{aligned}
 \text{Therefore, } Z^{-1}\{F(z)\} &= \frac{-ni^{n+1}}{4} + \frac{(-i)^{n+1}}{2} (1-n) \\
 &= \frac{-ni^{n+1}}{4} + \frac{(-i)^{n+1} i^{n+1}}{2} (1-n) \\
 &= \frac{i^{n+1}}{4} [-n + (-1)^{n+1} (2-2n)]
 \end{aligned}$$

$$(xxi) \frac{z^2 + z + 2}{(z-1)(z^2 - z + 1)}$$

Solution: Let,

$$\begin{aligned}
 F(z) &= \frac{z^2 + z + 2}{(z-1)(z^2 - z + 1)} \\
 &= \frac{z^2 + z + 2}{(z-1)\left(z^2 - 2 \cdot \frac{1}{2} \cdot z + \frac{1}{4} - \frac{1}{4} + 1\right)}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{z^2 + z + 2}{(z-1)\left(\left(z - \frac{1}{2}\right)^2 - \frac{3}{4}i^2\right)} \\
 &= \frac{z^2 + z + 2}{(z-1)\left(z - \frac{1}{2} - \frac{\sqrt{3}}{2}i\right)\left(z - \frac{1}{2} + \frac{\sqrt{3}}{2}i\right)}
 \end{aligned}$$

$$\text{Then, } z^{n-1} F(z) = \frac{z^{n+1} + z^n + 2z^{n-1}}{(z-1)\left(z - \frac{1}{2} - \frac{\sqrt{3}}{2}i\right)\left(z - \frac{1}{2} + \frac{\sqrt{3}}{2}i\right)}$$

Clearly $z^{n-1} F(z)$ has poles at $z = 1, \frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $\frac{1}{2} - \frac{\sqrt{3}}{2}i$ of simple order.

Here, residue of $z^{n-1} F(z)$ at $z = 1$ of order 1 is,

$$\text{Res}_{z=1} z^{n-1} F(z) = \lim_{z \rightarrow 1} \frac{z^{n-1} (z^2 + z + 2)}{z^2 - z + 1} = 1^{n-1} \frac{(1^2 + 1 + 2)}{1^2 - 1 + 1} = 4.$$

Also, residue of $z^{n-1} F(z)$ at $z = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ of order 1 be,

$$\begin{aligned}
 \text{Res}_{z=(1/2)+i(\sqrt{3}/2)} z^{n-1} F(z) &= \lim_{z \rightarrow \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)} \frac{z^{n-1} (z^2 + z + 2)}{(z-1)\left(z - \frac{1}{2} + \frac{\sqrt{3}}{2}i\right)} \\
 &= \frac{\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^{n-1} \left[\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^2 + \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) + 2\right]}{\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i - 1\right)\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i - \frac{1}{2} - \frac{\sqrt{3}}{2}i\right)}
 \end{aligned}$$

Also, residue of $F(z) z^{n-1}$ at $z = \frac{1}{2} - \frac{\sqrt{3}}{2}i$ of simple order be,

$$\begin{aligned}
 \text{Res}_{z=(1/2)-i(\sqrt{3}/2)} F(z) z^{n-1} &= \lim_{z \rightarrow \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)} \frac{z^{n-1} (z^2 + z + 2)}{(z-1)\left(z - \frac{1}{2} - \frac{\sqrt{3}}{2}i\right)} \\
 &= \frac{\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)^{n-1} \left[\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)^2 + \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) + 2\right]}{\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i - 1\right)\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i - \frac{1}{2} + \frac{\sqrt{3}}{2}i\right)}
 \end{aligned}$$

We know,

$$Z^{-1}\{F(z)\} = \text{sum of residues of } z^{n-1} F(z)$$